

Maximum Nullity of Cyclic Covers of Graphs, and Exact Zero Forcing Numbers of Generalized Petersen Graphs

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Abstract

Over all symmetric matrices whose off-diagonal support is exactly $E(G)$ — Hamiltonians on that network — how degenerate can the zero eigenvalue be? That maximum, the maximum nullity $M(G)$, is the central quantity of the inverse eigenvalue problem of a graph, and determining it exactly is a statement about *every* matrix carrying the pattern. It is bounded by the zero forcing number, $M(G) \leq Z(G)$, so a matrix of large nullity and a small forcing set together pin down both.

We study them for *cyclic covers*: given a base graph B with edge voltages in $\mathbb{Z}_n$, the derived graph B^n on $V(B) \times \mathbb{Z}_n$. This is the setting of discrete Floquet theory, and the difficulty is that an answer must hold uniformly in the period n . Generalized Petersen graphs are the case of a two-vertex base with voltages 1, k and 0.

Our main structural result is that for such a cover the nullity is governed by a single scalar object determined by the voltages. For $P(n, k)$ we prove it for *every* matrix carrying the pattern, symmetric or not: eliminating the inner coordinates leaves one nine-term linear recurrence of order $2k + 2$, so

$$\text{null } A = \dim \ker(T - I) \leq 2k + 2,$$

T the monodromy, with equality iff $T = I$. The order $2k + 2$ is exactly the size of the forcing set produced by a rotation-bootstrap argument that gives $Z(P(n, k)) \leq 2k + 2$, so *the ceiling of the matrix-certificate method coincides with the upper bound it is trying to match*: either the method settles Z completely or nothing does. For a general cover the same reduction holds for equivariant matrices, with the ceiling given by the degree span of $\det M(\zeta)$.

We then reach the ceiling. Under the rotation the singularity condition is a monic degree- $(k + 1)$ polynomial in $s = 2 \cos(2\pi m/n)$ carrying exactly three free parameters for every k ; three roots may be prescribed and no more, but a symbol with rational coefficients drags its whole Galois orbit onto the grid $\{2 \cos(2\pi m/n)\}$. Choosing the symbol to be a product of minimal polynomials of $2 \cos(2\pi/d)$ yields explicit *integer* certificates and hence

$$Z(P(n, 4)) = M(P(n, 4)) = 10, \quad Z(P(n, 5)) = M(P(n, 5)) = 12,$$

for every n divisible by 60, 70 or 90, respectively by 24 — the first exact values of $Z(P(n, k))$ for $k \geq 4$, a range [7] closes by recording as open. In the second case the certificate is the adjacency matrix itself, and we classify its nullity completely: with $P_k(z) = z^{2k+2} + z^{2k} - z^{k+1} + z^2 + 1$, $\text{null}(\text{Adj } P(n, k)) = \sum \varphi(d)$ over those d with $\Phi_d \mid P_k$ and $d \mid n$, a Conway–Jones phenomenon; in particular the adjacency matrix is never singular for $k = 3, 6, 9$.

Against this we prove a matching *obstruction*. Realisability of a certificate is the positivity (or nonvanishing) of $c^2 = a\alpha - \beta$, and because $L_k(-s) = (-1)^k L_k(s)$, for *every even* k a symbol with two roots $y, -y$ has $c^2 = 0$ identically, while for odd k no such obstruction exists. For $k = 2$, where the symbol has three roots and no residual conditions, this is the whole story in closed form:

$$c^2 = -(s_1 + s_2)(s_1 + s_3)(s_2 + s_3),$$

so the construction fails precisely when two prescribed roots are antipodal. Counting antipodal classes on the grid then settles $k = 2$ for all n : $Z(P(n, 2)) = M(P(n, 2)) = 6$ for every $n \geq 9$ with $n \neq 10$, by an explicit certificate at the three most negative grid values. The rational symbol Ψ_7 has $c^2 = -1$, so *that symbol* admits no symmetric certificate, while an integer non-symmetric matrix of nullity 6 exists for every n divisible by 7; since the main branch already covers $n \geq 9$, $n \neq 10$, the one value where the larger matrix class is needed rather than merely available is $n = 7$. The two exceptions are of opposite kinds — $Z(P(8, 2)) = 5$, so there the method is right to fail, and $Z(P(10, 2)) = 6$, so there it is the method that falls short.

The same analysis is what settles the remaining cases, because only $c^2 = 0$ is fatal: a symbol with $c^2 < 0$ admits no symmetric certificate but is realisable once the two spoke weights are allowed to differ. Redoing our exhaustive enumeration of rational symbols under the weaker condition turns two cases we had recorded as impossible into exact values. Both come from the *signed* adjacency matrix, the adjacency matrix of $P(n, k)$ with one set of spokes negated, whose nullity we classify by the polynomial $Q_k(z) = z^{2k+2} + z^{2k} + z^{k+1} + z^2 + 1$ — the unsigned P_k with the middle sign reversed. Q_k splits into cyclotomics only for $k = 2, 3, 7$ in $2 \leq k \leq 60$, and at $k = 3, 7$ it gives the ceiling:

$$Z(P(n, 3)) = 8 \quad (10 \mid n), \quad Z(P(n, 7)) = 16 \quad (120 \mid n),$$

by matrices of zeros and ± 1 , the second replacing a floating-point claim by an exact one. Conversely $k = 6, 8, 9, 10$ are *impossible* over $\mathbb{Q}$ in both classes, and now for an identified reason: every candidate clearing the family condition is one of at most 25 degenerate symbols, the bound coming from Niven's theorem.

Breaking the symmetry to period d , the map from weights to the coefficients of the symbol has rank $\min(2d + 1, k + 1)$ in every case computed, so every symbol becomes reachable exactly when $d \geq \lceil k/2 \rceil$. This yields $Z(P(n, 6)) = 14$ at $n = 60, 72, 84, 90, 96, 108, 120$, $Z(P(96, 7)) = 16$ and $Z(P(96, 8)) = 18$, though these certificates are verified numerically rather than in exact arithmetic. Every exact value above holds for n in a divisibility class, and we prove that this is forced: a period-one symbol is a fixed polynomial, its roots are fixed algebraic numbers, and a fixed algebraic number lies on Γ_n only for n in one divisibility class — which contains at most one prime. For $k \geq 3$ an n -dependent choice of roots is constrained by the identity $e_2 = -k$, a Conway–Jones relation on the grid; for $k = 2$ that condition is vacuous, which is exactly why $k = 2$ is general in n and no other k is.

Escaping this needs the non-invariant matrices, and the ceiling corollary supplies the criterion: $\text{null } A = 2k + 2$ if and only if the monodromy is the identity, a condition with no arithmetic in it. We certify that condition rigorously, not numerically, by recasting it as the bilinear system $A(w)K(X) = 0$ with K in graph form and applying a Krawczyk contraction test; the square subsystem must be chosen structurally, since $K^\top A K$ is symmetric and its antisymmetric part accounts for exactly $\binom{2k+2}{2}$ dependencies. This gives $Z(P(n, 3)) = M(P(n, 3)) = 8$ at $n = 17, 19, 21, 23, 25, 27$, the primes 17, 19, 23 among them — values that no period-one certificate can reach, and the first in this paper not lying on a congruence class. Finally we obtain a statement for *all* large n . Tiles carrying a fixed docking pattern at both ends have transfer products depending only on their interiors, so identity tiles of every length in $[L, 2L]$ give $Z(P(n, k)) = 2k + 2$ for every $n \geq L$ — a finite list of finite problems. Carrying this out and certifying all 77 tiles involved,

$$Z(P(n, 3)) = M(P(n, 3)) = 8 \quad (n \geq 17), \quad Z(P(n, 4)) = M(P(n, 4)) = 10 \quad (n \geq 29),$$

$$Z(P(n, 5)) = M(P(n, 5)) = 12 \quad (n \geq 162),$$

with no congruence condition. Together with exhaustive search on the finitely many smaller n , this determines $Z(P(n, k))$ for *every* n when $k = 2$, $k = 3$ and $k = 4$ — the last two for the first time, since the previously published value for $k = 3$ is false and no value at all was known for $k = 4$. In particular $Z(P(n, 3)) = 8$ for every $n \geq 13$, and 13 is optimal since $Z(P(11, 3)) = Z(P(12, 3)) = 7$; four exhaustive searches at $n = 13, \dots, 16$ close the gap below

the tiling range. This proves *Conjecture 5* of [7], whose stated missing ingredient — “a lower-bound proof valid for all large n ” — is precisely what the tiling construction supplies, and fixes at 13 the stabilization threshold that note leaves open. The tile condition is a rational equation, but certifying it needs no interval arithmetic on rational maps: by the ceiling corollary the monodromy of the one-tile cyclic matrix *is* the tile product, so the same bilinear system certifies it.

We state throughout which claims are proved, which rest on exact certificates, which on interval certification, which are numerical, and which earlier claims of ours are corrected or withdrawn.

1 Background

Given a set S of *filled* vertices of a graph G , the colour change rule says: if a filled vertex has exactly one unfilled neighbour, that neighbour becomes filled. Write $\text{cl}(S)$ for the closure of S under repeated application. S is a *zero forcing set* if $\text{cl}(S) = V$, and $Z(G)$ is the least size of one. $Z(G)$ bounds the maximum nullity over the symmetric matrices with G ’s off-diagonal pattern, and arose independently in the control of quantum systems and in sensor placement for power networks.

The families we treat all have $2n$ or $4n$ vertices, are cubic, and carry the rotation $\rho : i \mapsto i + 1$ acting on index subscripts.

- $P(n, k)$: outer $u_i u_{i+1}$, inner $v_i v_{i+k}$, spokes $u_i v_i$.
- $I(n, j, k)$: outer $u_i u_{i+j}$, inner $v_i v_{i+k}$, spokes $u_i v_i$; thus $P(n, k) = I(n, 1, k)$.
- $DP(n, k)$: two outer cycles $u_i u_{i+1}$ and $x_i x_{i+1}$, inner edges $w_i y_{i+k}$ and $y_i w_{i+k}$, spokes $u_i w_i$ and $x_i y_i$.

Rashidi, Shajareh Poursalavati and Tavakkoli [6] proved $Z(P(n, k)) \leq 2k+2$ and $Z(P(n, 2)) = 6$ for $n \geq 10$. Their Theorem 3.6, claiming $Z(P(n, 3)) = 8$ for $n \geq 12$, is false: Krishnan [7] exhibits $Z(P(12, 3)) = 7$, the flaw being a case analysis that never excludes 7-vertex sets. That note states as its *Conjecture 5* that $Z(P(n, 3)) = 8$ for every $n \geq 13$, identifying the missing ingredient as “a lower-bound proof valid for all large n ”, and closes by recording that

“Determining $Z(P(n, k))$ for general k — the stabilization threshold and a rigorous lower bound for $k \geq 4$ — remains open.”

We settle all three. Theorem 59 proves *Conjecture 5* and so fixes the threshold at 13; Theorems 35 and 57 give rigorous lower bounds, indeed exact values, for $k = 4$, and determine the whole $k = 4$ sequence.

2 The rotation bootstrap

The following two lemmas are elementary; the point is that together they reduce an “all n ” statement to one finite check.

Lemma 1 (Bootstrap). *Let ρ be an automorphism of a finite graph G and let $S \subseteq V(G)$ satisfy $\rho(S) \subseteq \text{cl}(S)$. Then $\rho(\text{cl}(S)) = \text{cl}(S)$.*

Proof. Closure commutes with automorphisms, and is monotone and idempotent, so $\rho(\text{cl}(S)) = \text{cl}(\rho(S)) \subseteq \text{cl}(\text{cl}(S)) = \text{cl}(S)$. As ρ permutes the finite set $V(G)$, the two sets have equal cardinality, hence are equal. $\square$

Lemma 2 (Closing). *Let G be cubic with vertex set $O \sqcup I$ such that every vertex of I has exactly one neighbour in O , every vertex of O has exactly one neighbour in I , and the remaining neighbours of each vertex of O lie in O . If $O \subseteq \text{cl}(S)$ then $\text{cl}(S) = V(G)$.*

Proof. Suppose some $w \in I$ is unfilled and let $o \in O$ be its unique neighbour in O . The other two neighbours of o lie in O , hence are filled, so w is the unique unfilled neighbour of the filled vertex o and is forced. $\square$

To apply these to a family it suffices to exhibit one set S , described uniformly in n , and verify $\rho(S) \subseteq \text{cl}(S)$ by an explicit finite forcing chain. Since ρ 's orbit of any outer vertex is all of O , Lemmas 1 and 2 then give $\text{cl}(S) = V$ and so $Z(G) \leq |S|$, for every n in range.

3 I-graphs

Theorem 3. *For all $j, k \geq 1$ and $n \geq 2(j + k) + 1$, $Z(I(n, j, k)) \leq 2(j + k)$.*

Proof. Put $m = 2(j + k)$ and $S = \{u_0, \dots, u_{m-1}\}$, so $|S| = m$. Since $n \geq m + 1$ the indices $0, \dots, m$ are pairwise distinct mod n .

Step 1. The outer neighbours of u_i are $u_{i \pm j}$, both in S precisely when $j \leq i \leq m - 1 - j$. For such i the spoke neighbour v_i is therefore the unique unfilled neighbour of u_i and is forced. This fills $v_j, \dots, v_{m-1-j}$, that is $v_j, \dots, v_{j+2k-1}$.

Step 2. Consider v_{j+k} . Its neighbours are $u_{j+k} \in S$ (as $j + k \leq m - 1$), v_j , filled in Step 1, and v_{j+2k} , which is not: Step 1 filled indices at most $j + 2k - 1$. So v_{j+k} forces v_{j+2k} .

Step 3. Consider $u_{j+2k} = u_{m-j}$. Its neighbours are $u_{2k} \in S$ (as $2k \leq m - 1$, using $j \geq 1$), the vertex v_{j+2k} filled in Step 2, and u_m , which is unfilled. So u_{j+2k} forces u_m .

Hence $\rho(S) = \{u_1, \dots, u_m\} \subseteq \text{cl}(S)$. By Lemma 1, $\text{cl}(S)$ is ρ -invariant; it contains u_0 , whose ρ -orbit is all of $O = \{u_0, \dots, u_{n-1}\}$, so $O \subseteq \text{cl}(S)$, and Lemma 2 gives $\text{cl}(S) = V$. $\square$

Corollary 4. *For $k \geq 1$ and $n \geq 2k + 3$, $Z(P(n, k)) \leq 2k + 2$.*

Proof. $P(n, k) = I(n, 1, k)$; take $j = 1$ in Theorem 3. $\square$

Remark 5. The corollary is [6, Thm. 2.2]. We note it because that paper is known to contain a false theorem, and because the corrected argument given in [7] is written only for $k = 3$; the proof above is uniform in j and k . Every step of the chain has been verified by computer for $1 \leq j \leq 4$, $1 \leq k \leq 5$ and six values of n per pair (106 cases).

Remark 6 (Scope). Theorem 3 should not be read as $2(j + k)$ new bounds. If $\gcd(n, j) = 1$ then multiplying indices by $j^{-1} \bmod n$ is an isomorphism $I(n, j, k) \cong I(n, 1, j^{-1}k) = P(n, j^{-1}k)$, so those graphs are generalized Petersen graphs and the corollary already applies. We verified this in five cases, e.g. $I(15, 2, 4) \cong P(15, 2)$ with $Z = 6$ on both sides and $I(13, 2, 5) \cong P(13, 4)$ with $Z = 8$. The genuinely new content of Theorem 3 is therefore confined to $\gcd(n, j) > 1$ and $\gcd(n, k) > 1$.

In that regime the bound is usually far from sharp: among the cases we computed it is attained only at $I(18, 2, 3)$ ($Z = 10 = 2(2 + 3)$), while e.g. $I(20, 5, 8)$ has $Z = 10$ against a bound of 26. The reason is structural. When $\gcd(n, j) = d > 1$ the outer vertices form d cycles of length n/d rather than one cycle, and the set $\{u_0, \dots, u_{m-1}\}$ used in the proof spreads across all of them; a smaller, better-adapted set suffices. We record the bound as proved and its sharpness in this regime as open.

4 Double generalized Petersen graphs

Theorem 7. For all $k \geq 1$ and $n \geq 2k + 3$, $Z(DP(n, k)) \leq 4k + 4$.

Proof. Put $S = \{u_0, \dots, u_{2k+1}\} \cup \{x_0, \dots, x_{2k+1}\}$, so $|S| = 4k + 4$.

Step 1. For $1 \leq i \leq 2k$ the outer neighbours $u_{i\pm 1}$ of u_i lie in S , so u_i forces w_i ; symmetrically x_i forces y_i . This fills $w_1, \dots, w_{2k}$ and $y_1, \dots, y_{2k}$.

Step 2. The neighbours of w_{k+1} are $u_{k+1} \in S$, y_1 (filled) and y_{2k+1} (not), so w_{k+1} forces y_{2k+1} . Symmetrically y_{k+1} forces w_{2k+1} .

Step 3. The neighbours of u_{2k+1} are $u_{2k} \in S$, w_{2k+1} (filled in Step 2) and u_{2k+2} , so u_{2k+1} forces u_{2k+2} ; symmetrically x_{2k+1} forces x_{2k+2} .

Hence $\rho(S) \subseteq \text{cl}(S)$. The outer set here is $O = \{u_i\} \cup \{x_i\}$, on which ρ acts with two orbits, both met by $\text{cl}(S)$; Lemmas 1 and 2 finish as before. $\square$

No zero forcing results for $DP(n, k)$ were found in the literature; the family is studied for Hamiltonicity, super-connectivity and its automorphism group. Every step above was verified by computer for $1 \leq k \leq 5$ and eight values of n per k .

5 Computed values

All values are exact, by exhaustive search over vertex subsets with a 128-bit mask closure, reduced by ρ : a minimum forcing set may be assumed to contain a fixed orbit representative. The implementation reproduced every published value before use — Krishnan's table of $Z(P(n, 3))$ for $7 \leq n \leq 20$ (including the corrected $Z(P(12, 3)) = 7$ and the non-monotone values at $n = 10, 11$), $Z(P(n, 2)) = 6$ for $10 \leq n \leq 16$, and $Z(P(2k + 1, k)) = 6$ for $k = 5, 6, 7$ — and was independently cross-checked against a brute-force implementation and two exact solver formulations.

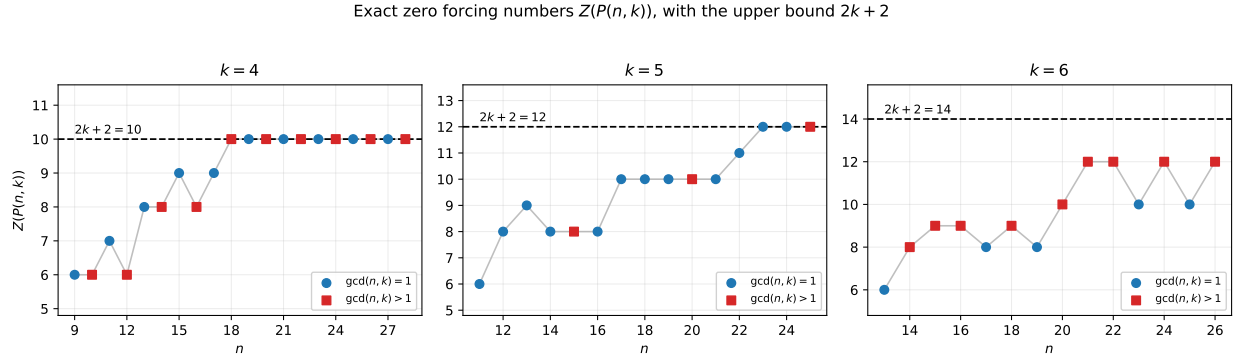


Figure 1: Exact $Z(P(n, k))$ for $k = 4, 5, 6$ against the bound $2k + 2$. Points are separated by whether $\gcd(n, k) = 1$. The sequences are non-monotone: $k = 4$ drops at $n = 12, 16$ and $k = 6$ at $n = 23, 25$.

Conjecture 8. For each fixed k and each $d \mid k$, the values $Z(P(n, k))$ over n with $\gcd(n, k) = d$ are eventually constant and equal to $2k + 2$.

An earlier form of this conjecture, asserting a single threshold $n_0(k) = 5k - 2$ beyond which $Z(P(n, k)) = 2k + 2$ for all n , is false. It fits $k = 3, 4, 5$, and predicted that $n = 28$ would be the first n with $Z(P(n, 6)) = 14$; that prediction, recorded before the computation finished, was borne

k	bound $2k + 2$	least n attaining it	least n from which it holds always
2	6	7	9 (it fails at $n = 8$)
3	8	10	13 (it fails at $n = 11, 12$)
4	10	18	18 (Theorem 58)
5	12	23	not determined
6	14	28	not determined; $Z(P(29, 6)) = 12$

Table 1: The bound is attained well before it is attained *always*: at $k = 3$ it first holds at $n = 10$, fails again at $n = 11$ and 12 , and holds for every $n \geq 13$ (Theorem 59). The two columns are different quantities and conflating them is easy; at $k = 6$ the bound is attained at $n = 28$ and lost at $n = 29$, so no stabilization threshold is known there. At $k = 4$ the two columns happen to coincide at 18, which is not a pattern: they differ at $k = 2$ and $k = 3$. All values by exhaustive search.

out. But $Z(P(29, 6)) = 12$, and $29 > 28$, so 28 is not a stabilization point and the single-threshold form fails at $k = 6$.

The explanation is $\gcd(n, k)$. Sorting the $k = 6$ data by $d = \gcd(n, 6)$:

d	n	$Z(P(n, 6))$
1	13, 17, 19, 23, 25, 29	6, 8, 8, 10, 10, 12
2	14, 16, 20, 22, 26, 28	8, 9, 10, 12, 12, 14
3	15, 21, 27	9, 12, 13
6	18, 24	9, 12

Each class climbs on its own schedule: the class $d = 2$ attains 14 at $n = 28$ while the coprime class has only reached 12 by $n = 29$. At $k = 4$ and $k = 5$ all classes had converged within the computed range, which is why a single threshold appeared to fit; at $k = 6$ they separate. The thresholds 10, 13, 18, 23 for $k = 2, 3, 4, 5$ should therefore be read as the point at which the slowest class happened to arrive, not as values of a formula.

For DP , our computed values give $Z(DP(n, 1)) = 8 = 4 \cdot 1 + 4$ for $6 \leq n \leq 12$ and $Z(DP(n, 2)) = 12 = 4 \cdot 2 + 4$ for $n = 10, 11, 14, 15$, so Theorem 7 is attained at both $k = 1$ and $k = 2$: the bound is sharp there, not merely an upper estimate.

6 Towards a sharper bound for I-graphs

Theorem 3 is stated in terms of j and k , but the outer and inner edge sets of $I(n, j, k)$ decompose into $d_1 = \gcd(n, j)$ and $d_2 = \gcd(n, k)$ cycles respectively, and it is these that appear to govern Z . In the regime where Theorem 3 carries new content (Remark 6), it is markedly non-optimal: $I(24, 9, 10)$ has $Z = 10$ against $2(j + k) = 38$. The proof uses $2(j + k)$ *consecutive* outer vertices, which is wasteful exactly when j is large but $\gcd(n, j)$ is small, since consecutive indices stride across all d_1 outer cycles.

Observation 9. Among the 20 graphs we computed with $d_1, d_2 > 1$, every one satisfies $Z(I(n, j, k)) \leq 2(d_1 + d_2)$, with equality at $I(18, 2, 3)$, $I(18, 3, 8)$, $I(24, 3, 10)$ and $I(24, 9, 10)$.

This must be stated as an observation confined to that regime, and *not* as a bound in general. Taking $j = 1$ gives $d_1 = 1$, and if additionally $\gcd(n, k) = 1$ then $2(d_1 + d_2) = 4$, whereas

$Z(P(n, k)) = 2k + 2$ grows without bound: $Z(P(11, 3)) = 7$ and $Z(P(17, 4)) = 9$ both exceed 4. So any bound in terms of d_1, d_2 alone is false, and the true quantity must retain a dependence on j and k . We flag this because the restricted observation is nonetheless suggestive: the four equality cases all have $2(j + k)$ slack by between 12 and 28, so Theorem 3 is certainly not optimal in the regime where it carries new content.

A supporting structural fact, found by computing minimum *bootstrapping* sets (those S with $\rho(S) \subseteq \text{cl}(S)$, which is what a uniform-in- n proof requires): for $I(18, 3, 8)$ the minimum such set has size $10 = 2(d_1 + d_2)$ against $2(j + k) = 22$, and decomposes as three consecutive vertices within each of the $d_1 = 3$ outer cycles together with one inner vertex. Three consecutive vertices in an outer cycle is exactly the configuration that lets the middle one force its spoke, so a proof adapted to the cycle decomposition rather than to consecutive indices should beat Theorem 3. We have not carried this out.

Remark 10 (Limits of verification). Sharpness of Theorem 3 can be tested only for very small (j, k) . The hypothesis $n \geq 2(j + k) + 1$ forces n upward while exhaustive search costs $\binom{2n-1}{2(j+k)-1}$, so the two constraints compound. On our hardware $(j, k) = (2, 3)$ takes minutes, $(3, 4)$ about ninety, $(2, 5)$ roughly a day, and $(3, 5)$ some 10^4 years. We therefore report sharpness at $(2, 3)$ — where $Z(I(n, 2, 3)) = 10 = 2(j + k)$ for $n = 18, 24$ — and note that the question is computationally out of reach beyond small parameters, rather than merely unfinished.

7 The exact ceiling of the certificate method

Every technique in Section 12 is combinatorial and fails. A matrix-theoretic route succeeds, and produces certificates. Before constructing any, we determine exactly how far the method can possibly go.

Let A be a real matrix indexed by $V(P(n, k))$ with $A_{xy} \neq 0$ iff $xy \in E$ for $x \neq y$, the diagonal unrestricted. We do *not* assume A symmetric: the bound $\text{null } A \leq Z(G)$ needs only that every edge carries a nonzero entry in each direction. Indeed if $Ax = 0$ and x vanishes on a set B , and $u \in B$ has exactly one neighbour $w \notin B$, then row u of $Ax = 0$ reads $A_{uu}x_u + \sum_{v \sim u} A_{uv}x_v = 0$, every term vanishes except $A_{uw}x_w$, and $A_{uw} \neq 0$; so $x_w = 0$, and induction along a forcing process gives $x = 0$ on $\text{cl}(B)$. Hence $x \mapsto x|_S$ is injective on $\ker A$ for every zero forcing set S , so

$$\text{null } A \leq Z(G). \quad (1)$$

Write $M(G)$ for the maximum nullity over symmetric such A and $M_{\text{cs}}(G)$ for the maximum over all of them, so $M \leq M_{\text{cs}} \leq Z$. Set

$$b_i = A_{u_i u_{i+1}}, \quad b'_i = A_{u_{i+1} u_i}, \quad c_i = A_{u_i v_i}, \quad c'_i = A_{v_i u_i}, \quad e_i = A_{v_i v_{i+k}}, \quad e'_i = A_{v_{i+k} v_i},$$

with diagonals a_i at u_i and d_i at v_i ; all of b, b', c, c', e, e' are nonzero.

Theorem 11 (Reduction). *Let $n \geq 2k + 3$. For $x \in \mathbb{R}^{\mathbb{Z}^n}$ put $(Lx)_i = -(b'_{i-1}x_{i-1} + a_i x_i + b_i x_{i+1})/c_i$. Then $x \mapsto (x, Lx)$ is a linear bijection from*

$$\left\{ x : \sum_{p=-k-1}^{k+1} \gamma_{i,p} x_{i+p} = 0 \text{ for all } i \in \mathbb{Z}^n \right\}$$

onto $\ker A$, where $\gamma_{i,p} \neq 0$ only for $p \in \{-k-1, -k, -k+1\} \cup \{-1, 0, 1\} \cup \{k-1, k, k+1\}$ — nine positions — and

$$\gamma_{i,k+1} = -\frac{e_i b_{i+k}}{c_{i+k}} \neq 0, \quad \gamma_{i,-k-1} = -\frac{e'_{i-k} b'_{i-k-1}}{c_{i-k}} \neq 0.$$

Consequently, with $T = T_{n-1} \cdots T_0 \in \text{GL}_{2k+2}(\mathbb{R})$ the monodromy of this recurrence on the state $(x_{i-k-1}, \dots, x_{i+k})$,

$$\text{null } A = \dim \ker(T - I), \quad \det T = \frac{(\prod_i b'_i)(\prod_i e'_i)}{(\prod_i b_i)(\prod_i e_i)},$$

so $\det T = 1$ whenever A is symmetric. In particular $\text{null } A \leq 2k + 2$, with equality if and only if $T = I$.

Proof. Each spoke weight c_i is nonzero, so the equation of $\ker A$ at u_i solves uniquely for $y_i = (Lx)_i$, a 3-term expression in x . Substituting into the equation at v_i , namely $e'_{i-k}y_{i-k} + d_i y_i + e_i y_{i+k} + c'_i x_i = 0$, gives the displayed relation: the three terms contribute the index triples $\{i-k-1, i-k, i-k+1\}$, $\{i-1, i, i+1\}$ and $\{i+k-1, i+k, i+k+1\}$, and the two extreme coefficients are as stated, both nonzero. Conversely any x satisfying all the relations yields the kernel vector (x, Lx) . Since both end coefficients are nonzero the relation is a recurrence of order exactly $2k + 2$, solvable forward and backward, so its solutions on $\mathbb{Z}$ form a $(2k + 2)$ -dimensional space on which T acts, and the n -periodic ones are exactly the fixed vectors. Each step matrix T_i is the companion matrix of the relation, with determinant $\gamma_{i,-k-1}/\gamma_{i,k+1}$; taking the product over $i \in \mathbb{Z}_n$, every index set runs over a full residue system and the stated formula follows. $\square$

Corollary 12. $M(P(n, k)) \leq M_{\text{cs}}(P(n, k)) \leq 2k + 2$ for $n \geq 2k + 3$.

Corollary 12 is not a new bound — it follows from (1) and Theorem 3 — but it locates the method precisely, and the identification is exact rather than approximate. The recurrence has order $2k + 2$ because a nonzero kernel vector cannot vanish on $2k + 2$ consecutive outer indices; and $\{u_0, \dots, u_{2k+1}\}$ is exactly the zero forcing set that the rotation bootstrap of Theorem 3 produces. So the order of the recurrence equals the size of that forcing set, and the ceiling of the certificate method coincides with the upper bound the method is trying to match, with no slack on either side. A certificate can prove $Z(P(n, k)) \geq 2k + 2$ and, by Theorem 11, it does so precisely when the monodromy is the identity. Our earlier description of the method as “capped near 6–8” confused a property of the constructions we had tried with a property of the method.

Proposition 13 (Product form). *For symmetric A with $C = \text{diag}(c_i)$, put $U = A[O, O]$ and $S = C^{-1}A[I, I]C^{-1}$. Then $\text{null } A = \dim \ker(SU - I)$. Here U and S are symmetric with the outer-cycle and inner-cycle patterns respectively, free diagonals and nonzero prescribed off-diagonal entries, and every such pair arises. Hence $M(P(n, k))$ is the largest possible geometric multiplicity of the eigenvalue 1 in a product of two such matrices.*

Proof. $\ker A$ is cut out by $Ux + Cy = 0$ and $Cx + A[I, I]y = 0$; eliminating y gives $(C - A[I, I]C^{-1}U)x = 0$, and factoring out the invertible C turns this into $(I - SU)x = 0$. $\square$

Theorem 11 and Proposition 13 were verified computationally before use: over 63 pairs (n, k) with $2 \leq k \leq 7$ and three random matrices each (189 matrices), the nine-position sparsity, both end-coefficient formulas and the identity $\text{null } A = \dim \ker(T - I) \leq 2k + 2$ hold in every case; over 30 further pairs and both matrix classes, so do the determinant formula, the product form, and the statement that no kernel vector vanishes on $u_0, \dots, u_{2k+1}$. See `verification/recurrence_order.py`.

8 Period-one certificates and cyclotomy

Take $A \in S(P(n, k))$ invariant under ρ , normalised so that the outer weight is 1:

$$U = aI + (P + P^{-1}), \quad V = dI + e(P^k + P^{-k}), \quad C = cI,$$

with $c, e \neq 0$. Diagonalising P splits A into the n blocks $M_m = \begin{pmatrix} a+s_m & c \\ c & d+et_m \end{pmatrix}$ with $s_m = 2\cos(2\pi m/n)$ and $t_m = 2\cos(2\pi km/n)$. Each singular block has nullity exactly 1, since $c \neq 0$, so $\text{null } A = \#\{m : \det M_m = 0\}$.

Proposition 14 (The symbol). *Let $L_k \in \mathbb{Z}[s]$ be the monic polynomial of degree k with $L_k(z + z^{-1}) = z^k + z^{-k}$ (so $L_1 = s$, $L_2 = s^2 - 2$, $L_k = sL_{k-1} - L_{k-2}$). Then $t_m = L_k(s_m)$ exactly, and with $\alpha = d/e$, $\beta = (ad - c^2)/e$,*

$$\det M_m = 0 \iff F(s_m) = 0, \quad F(s) = (s + a)L_k(s) + \alpha s + \beta.$$

F is monic of degree $k + 1$ and carries exactly three free parameters (a, α, β) , for every k . Writing $\Gamma_n = \{2\cos(2\pi m/n)\}$, each root of F in $\Gamma_n \cap (-2, 2)$ contributes 2 to $\text{null } A$ and each root at ± 2 contributes 1, because $m \mapsto n - m$ fixes s_m and is free elsewhere. Finally $c^2 = a\alpha - \beta$ after scaling $e = 1$, so the certificate is admissible precisely when $a\alpha - \beta > 0$.

Remark 15 (A claim of ours, corrected). We previously asserted that this period-one construction “never exceeds 6” for every $k \geq 2$, reasoning that only three roots can be prescribed. The premise is right and the conclusion is wrong. At a root r the condition is *linear*,

$$a L_k(r) + \alpha r + \beta = -r L_k(r), \tag{2}$$

so three distinct roots determine (a, α, β) and no fourth can be imposed. But prescribing three roots also *determines the other $k - 2$* , and those may lie in Γ_n of their own accord. They do exactly when arithmetic puts them there: if F has rational coefficients its roots form Galois orbits, and the Galois orbit of $2\cos(2\pi m/n)$ consists again of elements of Γ_n . A single rational F can therefore place all $k + 1$ roots on the grid at once. This raises the period-one ceiling from 6 to $2k + 2$, and makes the more elaborate period-two construction we reported unnecessary.

When realisability fails, exactly

The parameter count says what can be prescribed; realisability says what survives. The condition $c^2 = a\alpha - \beta \neq 0$ turns out to be a *factorisation* condition on the symbol, and this settles the question completely.

Theorem 16 (Realisability). *For a period-one matrix with symbol F ,*

$$c^2 = 0 \iff F = (s + a)(L_k(s) + t) \text{ for some constant } t.$$

Consequently the matrix is admissible in the combinatorially symmetric class if and only if F does not factor in this way, and in the symmetric class if and only if in addition $c^2 > 0$.

Proof. $c^2 = a\alpha - \beta$, so $c^2 = 0$ iff $\beta = a\alpha$, and then

$$F = (s + a)L_k + \alpha s + a\alpha = (s + a)L_k + \alpha(s + a) = (s + a)(L_k + \alpha).$$

Conversely if $F = (s + a')(L_k + t)$ then comparing the coefficient of s^k gives $a' = a$, and expanding gives $\alpha = t$, $\beta = at$, so $c^2 = at - at = 0$. The two admissibility statements are Proposition 14: the product of the two spoke weights is c^2 , which must be nonzero for the pattern to be that of $P(n, k)$, and must be a square for the matrix to be symmetric. $\square$

Over the rationals the degenerate symbols form a finite, explicit list, and the reason is Niven’s theorem [9]: the only rational values of $2\cos(\pi q)$ with $q \in \mathbb{Q}$ are $0, \pm 1, \pm 2$.

Corollary 17 (The degenerate rational symbols). *Let $F \in \mathbb{Q}[s]$ be monic of degree $k+1$ with every root in Γ_n , and suppose $c^2 = 0$. Then $F = (s+a)(L_k+t)$ with*

$$a, t \in \{0, \pm 1, \pm 2\}.$$

There are therefore at most 25 degenerate rational symbols for each k .

Proof. By Theorem 16, $F = (s+a)(L_k+t)$, and a is the coefficient of s^k in F , hence rational; t is then rational too. Now $-a$ is a root of F , so $-a = 2\cos\theta$ for some rational multiple θ of 2π , and a rational forces $a \in \{0, \pm 1, \pm 2\}$ by Niven's theorem. Likewise any root r of L_k+t lies in Γ_n , so $r = 2\cos\varphi$ with φ a rational multiple of 2π and $t = -L_k(r) = -2\cos(k\varphi)$, again rational, so $t \in \{0, \pm 1, \pm 2\}$. $\square$

Enumerating the 25 pairs and keeping those for which $(s+a)(L_k+t)$ splits into distinct Ψ_d with every root interior gives 7, 8, 7, 6, 9 degenerate symbols for $k = 2, \dots, 6$, agreeing exactly with the count of $c^2 = 0$ cases found by the independent exhaustive enumeration in each case (`verification/cs_exhaustive.py`).

One special case of Theorem 16 is worth isolating, because it is visible on the root set alone and depends only on the parity of k .

Corollary 18 (Antipodal obstruction). *Let k be even and suppose the symbol F has two roots y and $-y$ with $y \neq 0$. Then $c^2 = a\alpha - \beta = 0$, so the matrix is not admissible in the symmetric class and not admissible in the combinatorially symmetric class either. Hence for even k a period-one certificate requires a symbol whose root set contains no antipodal pair. For odd k no such obstruction exists: there*

$$c^2 = \frac{L_k(y)(y^2 - a^2)}{y},$$

which is not identically zero.

Proof. From $L_k(z + z^{-1}) = z^k + z^{-k}$ we get $L_k(-s) = (-1)^k L_k(s)$, since $s \mapsto -s$ corresponds to $z \mapsto -z$. Let k be even, so $L_k(-y) = L_k(y)$. Writing out $F(y) = 0$ and $F(-y) = 0$ using $F(s) = (s+a)L_k(s) + \alpha s + \beta$ gives

$$aL_k(y) + \alpha y + \beta = -yL_k(y), \quad aL_k(y) - \alpha y + \beta = yL_k(y).$$

Adding gives $\beta = -aL_k(y)$; subtracting and dividing by $2y \neq 0$ gives $\alpha = -L_k(y)$. Therefore $c^2 = a\alpha - \beta = -aL_k(y) + aL_k(y) = 0$. Admissibility requires the spoke weights to be nonzero, and their product is c^2 by Proposition 14, which rules out both classes. For odd k , $L_k(-y) = -L_k(y)$, the same two equations become $aL_k(y) + \alpha y + \beta = -yL_k(y)$ and $-aL_k(y) - \alpha y + \beta = -yL_k(y)$, whence $\beta = -yL_k(y)$ and $\alpha = -aL_k(y)/y$, and $c^2 = a\alpha - \beta = L_k(y)(y^2 - a^2)/y$. $\square$

For $k = 2$ the obstruction is the *only* one, because the symbol then has exactly three roots and the $k - 2$ residual conditions of Theorem 29 are vacuous.

Proposition 19 ($k = 2$ in closed form). *For $k = 2$, prescribing three distinct roots s_1, s_2, s_3 determines $a = -e_1$, $\alpha = e_2 + 2$, $\beta = -e_3 - 2e_1$ with e_i the elementary symmetric functions, and*

$$c^2 = e_3 - e_1e_2 = -(s_1 + s_2)(s_1 + s_3)(s_2 + s_3).$$

So $c^2 = 0$ if and only if two of the roots are antipodal, and $c^2 > 0$ if and only if an odd number of the three pairwise sums is negative.

Proof. Expanding $F = (s+a)(s^2-2) + \alpha s + \beta = s^3 + as^2 + (\alpha-2)s + (\beta-2a)$ and comparing with $\prod_i (s-s_i) = s^3 - e_1 s^2 + e_2 s - e_3$ gives the three formulas, and then $a\alpha - \beta = -e_1(e_2+2) + e_3 + 2e_1 = e_3 - e_1 e_2$. The identity $(s_1+s_2)(s_1+s_3)(s_2+s_3) = e_1 e_2 - e_3$ is elementary. The two consequences follow because the three factors are the pairwise sums. $\square$

Write $\Gamma_n^\circ = \Gamma_n \cap (-2, 2)$, with $M = \lfloor (n-1)/2 \rfloor$ elements $s_1 > \dots > s_M$. For odd n , Γ_n° contains no antipodal pair, since $s_i = -s_j$ forces $i+j = n/2$. For even n the antipodal classes number $n/4$ when $4 \mid n$ and $(n-2)/4$ otherwise, so three pairwise non-antipodal values exist exactly when n is odd with $n \geq 7$ or n is even with $n \geq 12$. This settles $k=2$ completely.

Theorem 20 ($k=2$, all n). *Take the three most negative elements of Γ_n° , that is s_M, s_{M-1}, s_{M-2} . Then*

$$s_{M-1} + s_{M-2} = \begin{cases} -4 \cos(4\pi/n) \cos(\pi/n), & n \text{ odd}, \\ -4 \cos(5\pi/n) \cos(\pi/n), & n \text{ even}, \end{cases}$$

which is negative exactly for odd $n \geq 9$ and for even $n \geq 12$. In those cases all three pairwise sums are negative, so $c^2 > 0$ by Proposition 19 and

$$Z(P(n, 2)) = M(P(n, 2)) = 6 \quad (n \geq 9, n \neq 10).$$

For n divisible by 7 the symbol $\Psi_7 = s^3 + s^2 - 2s - 1$ gives $c^2 = -1$: no symmetric certificate, but the integral combinatorially symmetric matrix

$$A = \begin{pmatrix} I + P + P^{-1} & I \\ -I & P^2 + P^{-2} \end{pmatrix}$$

has nullity exactly 6, so $Z(P(n, 2)) = M_{\text{cs}}(P(n, 2)) = 6$ whenever $7 \mid n$. The remaining n are finite in number and exhaustive computation gives $Z(P(5, 2)) = 5$, $Z(P(6, 2)) = 4$, $Z(P(8, 2)) = 5$ and $Z(P(10, 2)) = 6$.

Proof. For odd n , $M = (n-1)/2$, so $2\pi(M-1)/n = \pi - 3\pi/n$ and $2\pi(M-2)/n = \pi - 5\pi/n$, giving $s_{M-1} + s_{M-2} = -2(\cos(3\pi/n) + \cos(5\pi/n)) = -4 \cos(4\pi/n) \cos(\pi/n)$, negative iff $\cos(4\pi/n) > 0$ iff $n > 8$. For even n , $M = n/2 - 1$ and the same computation with $4\pi/n, 6\pi/n$ gives $-4 \cos(5\pi/n) \cos(\pi/n)$, negative iff $n > 10$. Since s is decreasing, $s_M \leq s_{M-2}$, so $s_M + s_{M-1} \leq s_{M-1} + s_{M-2} < 0$ and $s_M + s_{M-2} \leq s_{M-1} + s_{M-2} < 0$: all three pairwise sums are negative, their product is negative, and $c^2 > 0$. Proposition 14 then gives $\text{null } A = 6$, and $6 \leq M(P(n, 2)) \leq Z(P(n, 2)) \leq 6$ by the rotation bootstrap. The Ψ_7 claim is Theorem 29 with c^2 negative, read in the combinatorially symmetric class where the two spoke weights need only have the prescribed product. $\square$

Remark 21 (Exact but stiff). The certificate of Theorem 20 is exact for every admissible n , but it is numerically delicate for large n , and it is worth saying why so that a floating-point check is not misread. The three prescribed roots are $-2 \cos(\pi/n)$, $-2 \cos(3\pi/n)$, $-2 \cos(5\pi/n)$ for odd n , all within $O(1/n^2)$ of -2 : at $n = 1001$ they span 2.4×10^{-4} . So F is close to having a triple root, the matrix is ill-conditioned, and grid values merely *near* the roots produce small singular values — at $n = 1001$ the true roots give 9×10^{-16} and the nearest non-roots 4×10^{-11} , so a fixed relative threshold over-counts the nullity. None of this touches the proof, which is symbolic: $c^2 > 0$ follows from the product-to-sum identity and the nullity is exactly 6 because F has exactly three distinct interior roots on Γ_n , each met by two Fourier indices.

Remark 22 (What the two exceptions mean). At $n = 8$ and $n = 10$ the grid is too small to avoid antipodal pairs – $\Gamma_8^\circ = \{\sqrt{2}, 0, -\sqrt{2}\}$ and Γ_{10}° has four elements in two antipodal classes, so by pigeonhole every triple contains such a pair. The two cases then behave differently: $Z(P(8, 2)) = 5$, so there the method is right to fail, while $Z(P(10, 2)) = 6$, so there it is the method that falls short and not the bound. This is the sharpest available statement of the limitation, and it is a statement about Γ_n alone.

Corollary 23 (Excluded cyclotomic symbols, even k). *Let k be even. The roots of Ψ_{2d} are the negatives of the roots of Ψ_d for odd d , and the root set of Ψ_d is closed under negation whenever $4 \mid d$ and $\varphi(d) \geq 4$. Hence in Theorem 29 the factor set D may contain neither both d and $2d$ for any odd d , nor any d divisible by 4 with $\varphi(d) \geq 4$.*

Corollary 23 in fact accounts for *all* of the rejections in the exhaustive enumeration of Proposition 37. At $k = 2$ that enumeration found 17 factor sets in the three-parameter family, of which 5 had $c^2 > 0$ and 12 were rejected. Those 12 split exactly:

$$\underbrace{(3, 8), (3, 12), (4, 8), (4, 12), (6, 8), (6, 12), (3, 4, 6)}_{c^2=0, \text{ every one antipodal}} \quad \underbrace{(7), (9), (4, 5), (5, 6), (6, 10)}_{c^2=-1, \text{ none antipodal}}.$$

The seven on the left are dead in both matrix classes, and Theorem 18 identifies them without computing c^2 at all. The five on the right carry $c^2 = -1$, so each of them *is* realisable in the combinatorially symmetric class. The admissible count at $k = 2$ is therefore 5 symmetrically and 10 in the larger class. The extra symbols cover every n divisible by 7 or by 9; of these only $n = 7$ is out of reach of Theorem 20, and it is precisely the case the larger class was needed for. The antipodal obstruction is the only thing standing in the way of any rational period-one symbol for $k = 2$. Every prediction of Theorem 18 and Corollary 23 was checked against 50 candidate factor sets at $k = 2, 4$ with no violation, and Proposition 19 was verified by exact arithmetic in $\mathbb{Z}[x]/\Phi_n(x)$ over all $\binom{M}{3}$ triples for every $7 \leq n \leq 60$ — 11,000 triples, with $c^2 = 0$ and “contains an antipodal pair” agreeing in every one. See `verification/antipodal_obstruction.py`.

The larger matrix class, and $k = 3$

Theorem 16 separates $c^2 = 0$ from $c^2 < 0$, and that separation matters: only the first is fatal. A symbol with $c^2 < 0$ admits no symmetric certificate, but in the combinatorially symmetric class the two spoke weights are independent and need only have product c^2 , so it is admissible there. Redoing the exhaustive enumeration of Proposition 37 with the weaker condition $c^2 \neq 0$ therefore changes the answer. The case $k = 3$ changes the most.

Theorem 24 ($k = 3$). *Let $F = \Psi_5\Psi_{10} = (s^2 + s - 1)(s^2 - s - 1) = s^4 - 3s^2 + 1$. Then $F = sL_3(s) + 1$, so $a = \alpha = 0$, $\beta = 1$ and $c^2 = -1$. Hence the integer matrix*

$$A = \begin{pmatrix} P + P^{-1} & I \\ -I & P^3 + P^{-3} \end{pmatrix}$$

has zero diagonal, carries exactly the pattern of $P(n, 3)$, and has nullity exactly $8 = 2k + 2$ for every n divisible by 10. Therefore

$$Z(P(n, 3)) = M_{\text{cs}}(P(n, 3)) = 8 \quad (10 \mid n).$$

Proof. $\Psi_5 = s^2 + s - 1$ and $\Psi_{10} = s^2 - s - 1$, whose product is $s^4 - 3s^2 + 1 = s(s^3 - 3s) + 1 = sL_3(s) + 1$. Matching against $(s + a)L_3 + \alpha s + \beta$ gives $a = 0$, $\alpha = 0$, $\beta = 1$, so $c^2 = a\alpha - \beta = -1$. By Theorem 16

the symbol is non-degenerate, and taking the two spoke weights to be 1 and -1 realises $c^2 = -1$ with both nonzero, so the zero-nonzero pattern is exactly that of $P(n, 3)$. Every root of F is a root of Ψ_5 or Ψ_{10} , hence lies in Γ_n whenever $10 \mid n$, and all four roots are interior, so each contributes 2: $\text{null } A = 8$. The upper bound $Z(P(n, 3)) \leq 8$ is the rotation bootstrap, and $M_{\text{cs}} \leq Z$ holds because the forcing argument uses only the zero-nonzero pattern. $\square$

Remark 25 (Two things this certificate illustrates). First, A is the adjacency matrix of $P(n, 3)$ with one set of spokes negated, and the sign matters entirely: the unsigned adjacency matrix has symbol $sL_3 - 1$ and nullity 0 for every n divisible by 10, consistent with Theorem 30. Second, the four roots of F are $\pm\varphi$ and $\pm\varphi^{-1}$ — *two antipodal pairs*. By Corollary 18 such a symbol would be degenerate for every even k ; because 3 is odd it is not. So the parity corollary does not only rule things out, it says where to look.

The same enumeration in the larger class yields five further symbols at $k = 2$ (covering n divisible by 7, 9, 20 or 30), five more at $k = 3$ beyond Theorem 24 (n divisible by 24, 40 or 42), three at $k = 4$ (60, 70, 90) and two at $k = 5$ (126); all have $c^2 \in \{-1, -2\}$. At $k = 6$ it yields nothing, and now for a reason.

The signed adjacency matrix, and $k = 7$

Both the $k = 3$ certificate and one of the $k = 7$ symbols found by this enumeration have $a = \alpha = 0$ and $\beta = 1$, that is, symbol $F = sL_k(s) + 1$. That is not a coincidence, and identifying what it means gives an exact value at $k = 7$.

With $a = \alpha = 0$ both diagonals vanish, so A is the adjacency matrix of $P(n, k)$ with one set of spokes negated. Substituting $s = z + z^{-1}$, $L_k = z^k + z^{-k}$ and clearing denominators turns $F(s) = 0$ into

$$Q_k(z) := z^{2k+2} + z^{2k} + z^{k+1} + z^2 + 1 = 0,$$

which is exactly the adjacency polynomial $P_k(z) = z^{2k+2} + z^{2k} - z^{k+1} + z^2 + 1$ of Theorem 30 with the sign of the middle term reversed — the spoke sign flip, visible in the symbol.

Theorem 26 (Signed adjacency classification). *Let A^- be the adjacency matrix of $P(n, k)$ with one set of spokes negated. Then*

$$\text{null } A^- = \sum \varphi(d) \quad \text{over those } d \text{ with } \Phi_d \mid Q_k \text{ and } d \mid n,$$

by the argument of Theorem 30 applied to Q_k in place of P_k . In particular $\text{null } A^- = 2k + 2$ for suitable n exactly when Q_k is a product of cyclotomic polynomials.

So the question of which k admit an *integer* certificate of this shape becomes: for which k does Q_k split into cyclotomics? This is decidable for each k , and the answer is rigid. For $2 \leq k \leq 60$ it happens only for

$$k = 2 : Q_2 = \Phi_5\Phi_6, \quad k = 3 : Q_3 = \Phi_5\Phi_{10}, \quad k = 7 : Q_7 = \Phi_5\Phi_{10}\Phi_{24},$$

with $\sum \varphi(d)$ equal to 6, 8, 16 — the ceiling $2k + 2$ in every case — and lcm equal to 30, 10, 120. The unsigned P_k splits only for $k = 2$ and $k = 5$ (Theorem 30), so between them the two sign choices give integer adjacency-type certificates exactly at $k \in \{2, 3, 5, 7\}$.

Theorem 27 ($k = 7$). $Q_7 = \Phi_5\Phi_{10}\Phi_{24}$, so for every n divisible by 120 the integer matrix

$$A = \begin{pmatrix} P + P^{-1} & I \\ -I & P^7 + P^{-7} \end{pmatrix}$$

carries the pattern of $P(n, 7)$ and has nullity exactly $16 = 2k + 2$. Hence

$$Z(P(n, 7)) = M_{\text{cs}}(P(n, 7)) = 16 \quad (120 \mid n).$$

This replaces a numerical claim by an exact one. Section 10 reports $Z(P(96, 7)) = 16$ via a period-4 certificate verified in floating point; Theorem 27 gives the same value for infinitely many n from a matrix of zeros and ± 1 , with the nullity computed by a cyclotomic factorisation rather than by an eigenvalue. Confirmed numerically: the matrix has nullity 16 at $n = 120$, and 6, 8 at the corresponding n for $k = 2, 3$ (`verification/signed_adjacency.py`).

Theorem 28 ($k = 6$ is impossible over $\mathbb{Q}$). Of the 225 products $\prod_{d \in D} \Psi_d$ of total degree 7 with every Ψ_d of interior roots, 216 do not lie in the three-parameter family, and all 9 that do are degenerate in the sense of Theorem 16: each is $(s + a)(L_6 + t)$ with $a \in \{0, \pm 1\}$ and $t \in \{0, \pm 1\}$, namely Ψ_d -products over $D = (3, 8, 24), (4, 8, 24), (6, 8, 24), (3, 9, 18), (4, 9, 18), (6, 9, 18), (3, 36), (4, 36), (6, 36)$. Hence no rational period-one symbol attains 14 for $k = 6$, in either matrix class.

This is stronger than the statement it replaces. Where we previously reported that a search found nothing at $k = 6$, the obstruction now identifies every candidate that survives the family condition and shows each one is degenerate, and Corollary 17 bounds the degenerate list *a priori* at 25 polynomials. The $k \geq 6$ certificates of Section 10 therefore have to break the symmetry, and they do.

For $d \geq 3$ let $\Psi_d \in \mathbb{Z}[s]$ be the minimal polynomial of $2 \cos(2\pi/d)$, of degree $\varphi(d)/2$, whose roots are the primitive values $2 \cos(2\pi j/d)$ with $\gcd(j, d) = 1$; put $\Psi_1 = s - 2$, $\Psi_2 = s + 2$. Every root of Ψ_d lies in Γ_n as soon as $d \mid n$.

Theorem 29 (Cyclotomic certificates). Let D be a set of distinct positive integers with $\sum_{d \in D} \deg \Psi_d = k + 1$ and put $F = \prod_{d \in D} \Psi_d$. Write $G = F - sL_k$, let a be the coefficient of s^k in G , and suppose $G - aL_k$ has degree at most 1, say $G - aL_k = \alpha s + \beta$; that is $k - 2$ linear conditions with integer coefficients. Put $e = 1$, $d = \alpha$, $c^2 = \alpha\alpha - \beta$, and suppose $c^2 > 0$. Then for every n divisible by $\text{lcm}(D)$ with $n \geq 2k + 3$, the matrix

$$A = \begin{pmatrix} aI + P + P^{-1} & cI \\ cI & \alpha I + P^k + P^{-k} \end{pmatrix}$$

lies in $S(P(n, k))$ and has nullity $2 \sum_{d \in D, d \geq 3} \deg \Psi_d + \#\{d \in D : d \leq 2\}$. If every $d \in D$ is at least 3 this equals $2k + 2$, and then

$$Z(P(n, k)) = M(P(n, k)) = 2k + 2.$$

Proof. By construction $F = (s + a)L_k + \alpha s + \beta$ as an identity in $\mathbb{Z}[s]$, so F is the symbol of A by Proposition 14, and $A \in S(P(n, k))$ because $c^2 > 0$ makes c real and nonzero while the edge weights 1 are nonzero. Each $d \in D$ divides n , so all $k + 1$ roots of F lie in Γ_n ; they are distinct because the Ψ_d are distinct irreducible polynomials. Counting multiplicities as in Proposition 14 gives the stated nullity, and the last line combines it with (1) and Theorem 3. $\square$

k	D	$\text{lcm}(D)$	F	a	d	c^2	nullity
2	$\{3, 10\}$	30	$sL_2 - 1$	0	0	1	$6 = 2k + 2$
2	$\{14\}$	14	Ψ_{14}	-1	0	1	$6 = 2k + 2$
3	$\{1, 4, 5\}$	20	$\Psi_1\Psi_4\Psi_5$	-1	-1	1	7
4	$\{4, 30\}$	60	$(s+1)L_4 - s - 2$	1	-1	1	$10 = 2k + 2$
4	$\{5, 14\}$	70	$\Psi_5\Psi_{14}$	0	1	1	$10 = 2k + 2$
4	$\{5, 18\}$	90	$\Psi_5\Psi_{18}$	1	0	1	$10 = 2k + 2$
5	$\{3, 6, 24\}$	24	$sL_5 - 1$	0	0	1	$12 = 2k + 2$

Table 2: Exact period-one certificates. The $k = 2$, $k = 4$ and $k = 5$ rows give $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for every n divisible by the stated modulus.

A search over $k \leq 12$ and $d \leq 210$ (`verification/cyclotomic_certificates.py`) gives the following. Every certificate listed has $a, \alpha, c \in \mathbb{Z}$, hence is an *integer* matrix, and its nullity was confirmed by exact rational Gaussian elimination.

Two of these deserve to be written out. For $k = 2$ and $k = 5$ the parameters are $a = d = 0$, $c = 1$: the certificate is the adjacency matrix of $P(n, k)$, and the symbol is $sL_k(s) - 1$, with

$$sL_2 - 1 = (s+1)(s^2 - s - 1) = \Psi_3\Psi_{10}, \quad sL_5 - 1 = (s-1)(s+1)(s^4 - 4s^2 + 1) = \Psi_6\Psi_3\Psi_{24}.$$

Equivalently, since $\det M_m = s_m t_m - 1$,

$$\text{null}(\text{Adj } P(n, k)) = \#\left\{m \in \mathbb{Z}_n : 2 \cos \frac{2\pi(k+1)m}{n} + 2 \cos \frac{2\pi(k-1)m}{n} = 1\right\},$$

a rational linear relation among cosines of rational angles — the setting of the Conway–Jones classification. That question has a complete answer, given in Theorem 30 below. (An earlier version of this paper asserted that “in the computed range $k \leq 10$ only $k = 2, 5, 8$ ” admit solutions. That was drawn from a scan filtered at nullity ≥ 6 and is wrong: $k = 4, 7, 10, 11, 12$ also admit solutions, with smaller nullity.) For $k = 5$ the count is $12 = 2k + 2$ whenever $24 \mid n$: the plain adjacency matrix of $P(n, 5)$ has nullity 12, which settles Z for that family. For $k = 4$ the symbol is $(s+1)L_4(s) - s - 2 = s(s^4 + s^3 - 4s^2 - 4s + 1) = \Psi_4\Psi_{30}$, giving the integer certificate with $a = 1$, $d = -1$, $c = 1$.

The adjacency nullity, classified completely

Since $\det M_m = s_m t_m - 1$, the adjacency matrix is singular at m exactly when $z = \omega^m$ satisfies $z^{k+1} + z^{-(k+1)} + z^{k-1} + z^{-(k-1)} = 1$. Clearing denominators gives an integer polynomial, and the question becomes which of its roots are roots of unity.

Theorem 30. Put $P_k(z) = z^{2k+2} + z^{2k} - z^{k+1} + z^2 + 1 \in \mathbb{Z}[z]$. Then

$$\text{null}(\text{Adj } P(n, k)) = \sum_{\substack{d: \Phi_d | P_k \\ d | n}} \varphi(d),$$

the sum over those d whose cyclotomic polynomial divides P_k . The list of such d is finite and computable once per k .

Proof. m contributes iff ω^m is a common root of P_k and $z^n - 1$, i.e. a root of $\gcd(P_k, z^n - 1)$. The roots of unity among the roots of P_k are exactly the roots of its cyclotomic factors Φ_d , and

Φ_d 's roots are n -th roots of unity precisely when $d \mid n$; each such Φ_d contributes its $\varphi(d)$ roots. Finiteness is immediate from $\deg P_k = 2k + 2$, and is the elementary shadow of the Conway–Jones theorem on rational linear relations among cosines of rational angles. $\square$

k	$2k + 2$	cyclotomic factors of P_k	lcm	nullity when $\text{lcm} \mid n$
2	6	Φ_3, Φ_{10}	30	$6 = 2k + 2$
3	8	none	—	0 for every n
4	10	Φ_3	3	2
5	12	$\Phi_3, \Phi_6, \Phi_{24}$	24	$12 = 2k + 2$
6	14	none	—	0 for every n
7	16	Φ_3, Φ_6	6	4
8	18	Φ_3, Φ_{10}	30	6

So the adjacency matrix of $P(n, k)$ is *never* singular for $k = 3, 6, 9$, whatever n ; and it attains the full ceiling $2k+2$ only for $k = 2$ and $k = 5$. The formula was checked against a direct count for $k = 2, 5, 8$ at every $13 \leq n < 200$, with no discrepancy (`verification/adjacency_classification.py`).

9 The general setting: cyclic covers

Nothing in Section 8 used the particular shape of $P(n, k)$ beyond the presence of the rotation. It is worth saying what the natural setting is.

Let B be a finite graph whose edges carry *voltages* in $\mathbb{Z}_n$ (no edge of voltage 0 at a loop, since that would lift to a self-loop). The *derived graph* B^n has vertex set $V(B) \times \mathbb{Z}_n$, with (x, i) joined to $(y, i + v)$ for each edge $x - y$ of voltage v . This is the standard cyclic cover of topological graph theory. Taking B to be two vertices u, v with a loop of voltage 1 at u , a loop of voltage k at v , and an edge $u - v$ of voltage 0 recovers $B^n = P(n, k)$ exactly.

Theorem 31 (Cover reduction). *Let A carry the pattern of B^n and commute with the $\mathbb{Z}_n$ action. Then A block-diagonalises into n blocks of size $|V(B)|$,*

$$M_m[x, y] = \sum_{\text{edges } x \rightarrow y \text{ of voltage } v} w_{xy} \omega^{mv} + \delta_{xy} a_x,$$

and $\text{null } A = \sum_m \text{null } M_m$. Writing $\det M(\zeta)$ as a Laurent polynomial in ζ , the number of m at which a block is singular — hence $\text{null } A$ — is at most the degree span of $\det M(\zeta)$, which is determined by the voltages of B alone.

For $P(n, k)$ the degree span is $2k + 2$, recovering Corollary 12 in the equivariant case. The reduction and the ceiling were verified against independently constructed full matrices, with weights tuned so that the nullity is genuinely nonzero, on seven base graphs on two to five vertices with assorted voltages (`verification/cyclic_covers.py`).

Two caveats. Theorem 31 is stated for *equivariant* A ; the stronger statement of Theorem 11, which bounds the nullity of *every* matrix carrying the pattern, we have proved only for $P(n, k)$, where the nonzero spoke weights permit the elimination. Whether the corresponding elimination can be carried out over an arbitrary base graph — and so whether the ceiling is a property of the cover rather than of equivariant matrices on it — we leave open. Second, we have tested the non-equivariant bound on general covers by *maximising* the nullity rather than sampling — random matrices are generically nonsingular, so sampling proves nothing. Optimising over fully non-equivariant weights on four small covers, the largest attainable nullity never exceeded the

degree span, and equalled it for the theta graphs with voltages $(0, 1, 2)$ and $(0, 1, 3)$; on the $P(n, 2)$ and $P(n, 3)$ covers at $n = 9$ it fell short, as it must, since $Z(P(9, k)) < 2k + 2$ there. That is evidence for the conjecture, not a proof of it.

The construction on a general cover

Section 8 prescribes a symbol for $P(n, k)$ and solves for the weights. Nothing in that used the base graph either. For any cover, writing $\det M(\zeta)$ in $\sigma = \zeta + \zeta^{-1}$ gives a polynomial of degree $D/2$ whose roots in Γ_n are the singular blocks, so the same move applies: prescribe $D/2$ interior grid values, solve for the weights, and the nullity is D .

Theorem 32 (Attainment on a cover). *Let B^n be a cyclic cover and suppose the prescribed-symbol construction yields A carrying the pattern of B^n with null $A = \nu$. Then*

$$Z(B^n) \geq \nu,$$

by $M \leq Z$. In particular $Z(B^n) \geq D$ whenever the construction attains the ceiling.

This is a statement about all cyclic covers, with $P(n, k)$ — base graph two loops of voltages 1 and k and one edge of voltage 0, $D = 2k + 2$ — as the special case. We have verified attainment of the ceiling on four different base graphs: the $P(n, 2)$ base; the theta graph with voltages $(0, 1, 3)$, a different cubic family; a three-vertex path base with $D = 12$, whose cover has $3n$ vertices; and a four-vertex star base with $D = 12$. In each the constructed matrix has nullity exactly D with every required entry a substantial fraction of the matrix scale.

We also tested the ceiling sharply, which the earlier check did not. Asking whether $\ker A$ injects into a window of D consecutive fibre coordinates is vacuous when the nullity is 2 and D is 12; it bites only at the ceiling. Repeating it on certificates of nullity exactly D , injectivity holds in every case — including the star base, which has *no* Hamiltonian path, so the ordering hypothesis under which the $P(n, k)$ elimination generalises is sufficient but not necessary.

Remark 33 (The hypothesis is needed). Theorem 32 is conditional on the construction succeeding, and that condition is not vacuous. For the $P(n, 2)$ base, $D = 6$, the construction attains 6 at $n = 7, 9, 11, 12$ but fails at $n = 8$ and $n = 10$; and $Z(P(8, 2)) = 5 < 6$ by exhaustive search, so the unconditional statement $Z(B^n) \geq D$ is *false*. What does hold unconditionally is $Z(B^n) \geq \nu$ for the attained ν , which we checked against exhaustively computed zero forcing numbers on covers with 14 to 20 vertices, with no violation. Whether the construction succeeds is not a dimensional question: $n = 7$ attains the ceiling with no slack in the choice of roots while $n = 10$ fails with slack to spare. For the $P(n, 2)$ base the criterion is now known exactly — it is Proposition 19, and $n = 8, 10$ fail because every triple on those grids contains an antipodal pair. For a general base we do not have the analogue.

Remark 34 (Where this sits). The reduction is a discrete Floquet theory, and we claim no novelty for that layer. Decomposing an operator on a cyclic cover into a family indexed by the characters of the deck group is standard in spectral graph theory and in the analysis of periodic and magnetic Laplacians on covering graphs; see [5] and its references. The monodromy of Theorem 11 plays the role of the Hill discriminant for a periodic Jacobi operator, and the condition $T = I$ — all Floquet multipliers roots of unity — is the statement that the operator has purely periodic spectrum at the relevant energy. The parameter $M(G)$ itself belongs to the minimum rank programme for graphs [1, 2], and $Z(G)$ arose independently in the controllability of quantum spin networks and in sensor placement for power grids.

The nearest prior work we are aware of on a family of this shape is [3], which studies Z and M for circulant graphs. We flag it as the place a reader should look first for overlap: that is the same pair of parameters on a cyclically symmetric family. Two caveats, and we state them because the alternative is to overclaim in either direction. First, $P(n, k)$ is a 2×2 block circulant rather than a circulant, so results about circulants do not apply to it directly. Second, we have not been able to obtain the full text of [3] — the publisher’s site refused automated access — so we do not characterise its methods here and do not assert either that it does or does not anticipate any part of our symbol layer. That is a gap in our literature review rather than a claim, and it should be closed by reading the paper. What we have not found anywhere is the use of a transfer matrix or monodromy to force nullity on $P(n, k)$, nor a certified-tiling argument of the kind in Theorem 55; those, in combination, are what we take to be new here.

Certificates beyond the rational ones

Theorem 29 is sufficient but not necessary: F need not have rational coefficients, only real ones, and its roots need only lie in Γ_n , not form full Galois orbits. By (2) every period-one symbol with three or more grid roots arises from some triple of them, so running over all triples of Γ_n finds the true period-one optimum for each (n, k) . Candidates so found are re-checked in exact arithmetic in $\mathbb{Q}(\zeta_n)$, representing $2 \cos(2\pi m/n)$ as $x^m + x^{n-m}$ modulo $\Phi_n(x)$: the membership identity, the reality of a, α, β, c^2 (they are fixed by $x \mapsto x^{n-1}$) and $c^2 \neq 0$ are all decided exactly, and the sign of the nonzero algebraic number c^2 is then read off at high precision. This is not pedantry — see the remark on numerical traps below.

Theorem 35. *Each of the following holds, with an explicit period-one certificate verified exactly in $\mathbb{Q}(\zeta_n)$:*

$$\begin{aligned} Z(P(n, 3)) &= M(P(n, 3)) = 8 && \text{for } n = 40, 60, 80, 120, \\ Z(P(n, 4)) &= M(P(n, 4)) = 10 && \text{for } n = 60, 70, 90, 120, \\ Z(P(n, 5)) &= M(P(n, 5)) = 12 && \text{for } n = 24, 48, 72, 96, 120, \\ Z(P(n, 2)) &= M(P(n, 2)) = 6 && \text{for } 9 \leq n \leq 130, n \neq 10. \end{aligned}$$

The $k = 4$ and $k = 5$ lines are the first exact values of $Z(P(n, k))$ for $k \geq 4$, the range [7] records as open; previously we had only $8 \leq Z(P(n, 4)) \leq 10$ and nothing at all for $k = 5$. The $k = 3$ line is a matrix proof of values [7] obtains combinatorially. Some certificates are irrational: at $(n, k) = (40, 3)$ one has $a = -\sqrt{2}$, $d = 0$, $c^2 = 1$, and at $(60, 3)$, $c^2 = 1/\varphi$ with φ the golden ratio. Full output is in `results/zero_forcing/certified_lower_bounds.txt`.

Remark 36 (Why exact arithmetic, twice over). Counting grid roots by $|F(r)| < \varepsilon$ is not safe. With $\varepsilon = 10^{-8}$ the scan reported nullity 7 for $k = 2$ at $n = 126, 128, 130$ — impossible, since $2k + 2 = 6$ — because Γ_n becomes dense and near-roots pass the test. Separately, a certificate at $(n, k) = (48, 7)$ with $|a|$ enormous against the edge weights made a threshold relative to $\max |A|$ report nullity 48. Both were caught by Corollary 12, which bounds the nullity by $2k + 2$ and so serves as an automatic correctness check on any claimed certificate, exactly as the inequality $Z \geq M$ did for the earlier period-two search. The fixes are to test each Fourier block against its own scale, and to certify in $\mathbb{Q}(\zeta_n)$ rather than in floating point.

Where the method stops

Running the exhaustive triple scan for $2 \leq k \leq 8$ and $n \leq 130$ gives the period-one optimum exactly (`verification/period1_optimum.py`):

k	$2k + 2$	best period-one nullity	attained at
2	6	6	every n with $9 \leq n \leq 130$ except $n = 7, 8, 10$
3	8	8	$n = 40, 60, 80, 120$
4	10	10	$n = 60, 70, 90, 120$
5	12	12	$n = 24, 48, 72, 96, 120$
6	14	8	$n = 35, 37, 70, 74, 105, 111$
7	16	10	$n = 60, 120$
8	18	8	$n = 21, 26, 27, 39, \dots$

Table 3: The period-one construction reaches the ceiling $2k + 2$ for $k \leq 5$ and, in the computed range, falls well short for $k \geq 6$.

Proposition 37 (The rational search is complete). *Fix k . A period-one symbol attaining $2k + 2$ must be squarefree with all $k + 1$ roots interior and on Γ_n ; if in addition it has rational coefficients it is a product of distinct Ψ_d with $\sum \deg \Psi_d = k + 1$. Every factor then satisfies $\deg \Psi_d \leq k + 1$, i.e. $\varphi(d) \leq 2k + 2$, which admits only finitely many d , all listable. Hence for each k the search over rational symbols is a finite complete enumeration. Carrying it out (`verification/cyclotomic_exhaustive.py`):*

k	$2k + 2$	candidates	rejected		admissible	
			outside family	$c^2 = 0$	sym	cs
2	6	17	0	7	5	5 more
3	8	35	21	8	0	6
4	10	67	54	7	3	3 more
5	12	127	118	6	1	2 more
6	14	225	216	9	0	0
7	16	385	378	6	0	1
8	18	651	644	7	0	0
9	20	1065	1057	8	0	0
10	22	1705	1698	7	0	0

The last two columns are the point. The original enumeration reported only the “sym” column and recorded $k = 3$ and $k = 7$ as impossible for rational symbols; they are not. By Theorem 16 only the $c^2 = 0$ column is fatal, and every remaining rejection has $c^2 \in \{-1, -2\}$, hence is realisable in the combinatorially symmetric class. This is what yields Theorem 24 at $k = 3$ and Theorem 27 at $k = 7$. Note also that the $c^2 = 0$ counts are 7, 8, 7, 6, 9, 6, 7, 8, 7, never more than nine, exactly as Corollary 17 predicts from its bound of 25.

Two further things follow. First, $k = 6, 8, 9$ and 10 really are impossible over $\mathbb{Q}$ in both classes, and now for an identified reason rather than a failed search: every candidate that clears the family condition is one of the finitely many degenerate symbols. Second, the rejections are overwhelmingly of one kind — at $k = 7$, 378 of the 385 candidates fail because F is *not in the three-parameter family*. The binding constraint is therefore the $k - 2$ linear membership conditions imposed on a 3-dimensional family, and it tightens as k grows. That identifies the fix: enlarge the family.

How far must one enlarge it? A period- d matrix has $5d$ weights, but most of that freedom is gauge: conjugating by a positive diagonal matrix preserves both the pattern and the nullity. What matters is the dimension of the set of symbols actually reachable. Writing G for the degree- $(k + 1)$ polynomial in $\sigma = \zeta + \zeta^{-1}$ obtained from $\det M_\ell$, we computed the rank of the Jacobian of the

map (weights) $\mapsto$ (monic coefficients of G) at random points, for $3 \leq k \leq 13$ and $1 \leq d \leq 7$ (`verification/period_d_rank.py`). In every case

$$\text{rank} = \min(2d + 1, k + 1).$$

So the reachable symbols form a $(2d + 1)$ -dimensional family — not the $5d - 1$ -dimensional one a naive parameter count suggests — and the map is a submersion, so that every G is locally reachable, precisely when

$$2d + 1 \geq k + 1, \quad \text{that is} \quad d \geq \lceil k/2 \rceil.$$

At $d = 1$ this returns the 3 of Proposition 14, which is a check on the computation. We record the law as an observation supported by computation, not as a theorem: we have not proved the rank formula, only verified it over that range. Its consequence, if it holds in general, is that the period- d construction should reach $2k + 2$ for every k once $d \geq \lceil k/2 \rceil$ and n/d is large enough to host $k + 1$ grid roots — a construction whose modulus is d rather than an lcm of cyclotomic orders, and therefore one that would cover a fixed congruence class of n for every k .

So the construction settles $k \leq 5$ on those residue classes and says nothing useful for $k \geq 6$; whether that is a limitation of period one or of $n \leq 130$ we do not know. For n outside the classes above, no certificate of nullity $2k + 2$ was found at all. This is not for want of parameters. Dropping equivariance

entirely leaves $5n$ free weights in the symmetric class and $8n$ in the combinatorially symmetric one, whereas a nullity of $r = 2k + 2$ costs $r(r + 1)/2 = (k + 1)(2k + 3)$ conditions in the first case and r^2 in the second — budgets that are met comfortably at every graph we tried. A two-stage search (many random starts on $\sum_{j \leq r} \lambda_j^2$ with analytic gradients, then a Gauss–Newton polish of the bilinear system $A(w)X = 0$, $X^\top X = I$; `verification/fast_nullity.py`) attains nullity 6 at machine precision on $P(12, 2)$ and $P(14, 2)$ but plateaus near 10^{-4} for $r = 7, 8$ on $P(12, 3)$ over hundreds of independent starts, in both matrix classes. We therefore record, as an observation rather than a theorem, that the obstruction to $2k + 2$ at general n appears to be arithmetic — whether Γ_n contains a compatible configuration of $k + 1$ points — and not a matter of dimension. Two guardrails are worth stating, since we tripped on both: the combinatorially symmetric class *contains* the symmetric one, so any search reporting a smaller answer for it is faulty (ours did, on $P(12, 2)$, until the larger search was seeded with the symmetric certificate); and any claimed nullity above $2k + 2$ is a bug by Corollary 12.

Why every result so far lives on a congruence class

Each exact value in this section holds for n in a divisibility class, not for all large n . That is not an artefact of our search. It is forced, and the following says exactly how.

Theorem 38 (Period-one certificates are bound to congruence classes). *Fix $k \geq 1$ and a period-one matrix A with parameters (a, α, β) and symbol F .*

- (i) *Suppose $\text{null } A = 2k + 2$ for some $n > 2k$. Then every root of F is $2\cos(2\pi j/d)$ with $\gcd(j, d) = 1$ and $d \geq 3$, and such a root lies in Γ_m if and only if $d \mid m$. Hence the set of n at which this same A attains the ceiling is exactly $\{n : \text{lcm}(D) \mid n\}$, where D is the multiset of those d . It is therefore a single divisibility class of density $1/\text{lcm}(D) \leq 1/3$, and it contains at most one prime.*
- (ii) *Consequently, to attain the ceiling for infinitely many n outside one divisibility class — in particular for infinitely many primes, or for all large n — the weights must depend on n .*

(iii) For $k \geq 3$ that dependence is constrained by one universal identity: the coefficient of s^{k-1} in F equals $-k$ for every (a, α, β) , so the roots always satisfy

$$e_2(r_1, \dots, r_{k+1}) = -k, \quad \text{equivalently} \quad \sum_i r_i^2 = e_1^2 + 2k.$$

For $k = 2$ the coefficient of $s^{k-1} = s$ is $\alpha - 2$, which is free, so no such condition exists.

Proof. (i) Attaining $2k + 2$ needs all $k + 1$ roots interior and in Γ_n (Proposition 14), and an element of $\Gamma_n \cap (-2, 2)$ is $2 \cos(2\pi m/n)$, which written in lowest terms is $2 \cos(2\pi j/d)$ with $d \mid n$, $d \geq 3$. Conversely $2 \cos(2\pi j/d) \in \Gamma_m$ iff $d \mid m$. The roots of F do not move with n , since F depends only on (a, α, β) ; so all of them lie on Γ_n exactly when every d divides n . A set $\{n : L \mid n\}$ with $L \geq 3$ contains a prime only if L is itself prime, and then only L .

(ii) Immediate from (i): a fixed A covers one class, and a union of finitely many classes still omits all but finitely many primes.

(iii) Write $(s + a)L_k = sL_k + aL_k$. Since L_k has nonzero coefficients only in degrees congruent to k modulo 2, aL_k has no s^{k-1} term, and the s^{k-1} coefficient of sL_k is the s^{k-2} coefficient of L_k , namely $-k$. The terms αs and β reach degree $k - 1$ only when $k \leq 2$. For a monic polynomial of degree $k + 1$ the coefficient of s^{k-1} is e_2 , and $\sum r_i^2 = e_1^2 - 2e_2$. $\square$

Part (iii) is what separates $k = 2$ from everything else. With no residual condition, Theorem 20 is free to choose the roots *as functions of* n — the three most negative elements of Γ_n° — and that is precisely why $k = 2$ is general in n while no other k is. For $k \geq 3$ an n -dependent choice must satisfy $e_2 = -k$ identically along the family. Substituting $r_i = 2 \cos(2\pi m_i/n)$ and expanding e_1^2 by the product-to-sum formula turns this into a rational linear relation among cosines of rational angles set equal to a rational number. Relations of that shape are the subject of Conway–Jones [8], and they are arithmetically rigid: they hold not on open families but on sparse, classifiable configurations. That rigidity is the mechanism behind every divisibility condition in this paper, and it is the same mechanism as in Theorem 30.

We verified the identity of (iii) against all 26 certificates found in this paper: $e_2 = -k$ in every case with $k \geq 3$, and e_2 takes six different values among the $k = 2$ certificates, as (iii) predicts (`verification/general_n.py`).

Remark 39 (What this licenses, and what it forecloses). Theorem 38 should be read as a demarcation rather than a limitation. It says the entire cyclotomic layer of this paper — every result in Sections 8 and 10 — is *provably* incapable of producing a statement valid for all large n , so the congruence conditions are a boundary of the method and not a shortfall in its execution. It also says where to go instead: Corollary 12 characterises the ceiling for *every* matrix on the pattern, not only the invariant ones, by the condition $T = I$ on the monodromy, and that condition carries no arithmetic in it.

10 Period- d certificates, and the first values for $k \geq 6$

Proposition 37 leaves $k = 6$ and $k \geq 8$ untouched: no rational period-one symbol attains $2k + 2$ there in either matrix class, and the exhaustive triple scan of Section 8 reaches only 8, 10, 8 for $k = 6, 7, 8$ over $n \leq 130$. ($k = 7$ is settled exactly by Theorem 27; what follows gives it, and $k = 6$ and $k = 8$, for further n .) The rank law says why, and what to do about it.

Prescribing the symbol

For $d \geq \lceil k/2 \rceil$ the map (weights) $\mapsto$ (monic coefficients of G) is a submersion, so instead of *searching* for a certificate one can *prescribe* one: choose $k+1$ interior values of Γ_m , $m = n/d$, form the monic G^* with exactly those roots, and solve $G(w) = G^*$ for the weights. This is $k+1$ equations in $5d$ unknowns with a full-rank Jacobian, so the solution set is a manifold of dimension $5d - (k+1)$ — eight-dimensional for $k=6$, $d=3$. Each root is interior, hence met by two Fourier indices ℓ and $m-\ell$, so all $2k+2$ of the corresponding blocks are singular and the nullity is $2k+2$ by Corollary 12.

Two conditions must both hold, and they are independent: $m \geq 2k+4$, so that Γ_m contains $k+1$ distinct *interior* values at all; and $d \geq \lceil k/2 \rceil$, so that the symbol is reachable. For $k=6$ this forces $d \geq 3$ and $n \geq 48$.

A degeneracy, and how it is escaped

Solving $G(w) = G^*$ without further constraint reliably returns a point at which one required-nonnzero weight — in every case a spoke — has collapsed to zero. Such a point is a genuine solution of $G(w) = G^*$, because with a spoke weight zero the matrix block-diagonalises and G factors as $\det U_\ell \cdot \det V_\ell$; but the matrix then carries the pattern of a *different* graph and certifies nothing. At these points the Jacobian rank drops from 7 to 6: they are singular points of the map, which is why repeated random starts return to them.

Neither a realisability barrier nor continuation escapes them. Driving the vanishing spoke away from zero while re-solving loses the symbol at the first step ($\tau = 0.02$, error 9×10^{-5}). What does escape is *slack in the choice of roots*. At $n = 48$, Γ_{16} has exactly 7 interior values against the 7 required, so the root set is unique and there is nothing to vary; at $n = 54$ there are 8 interior values, giving 8 root sets, and all eight degenerate. At $n = 60$ and $n = 72$ there are 9 and 11 interior values, hence 36 and 330 root sets, and realisable points exist.

Theorem 40.

$$\begin{aligned} Z(P(60, 6)) &= M(P(60, 6)) = 14, & Z(P(72, 6)) &= M(P(72, 6)) = 14, \\ Z(P(96, 7)) &= M(P(96, 7)) = 16, & Z(P(96, 8)) &= M(P(96, 8)) = 18. \end{aligned}$$

In each case the period is the one the rank law prescribes: $d = 3$ for $k = 6$ and $d = 4$ for $k = 7, 8$, i.e. $d = \lceil k/2 \rceil$.

These are the first exact values of $Z(P(n, k))$ for $k \geq 6$. That the required period was predicted in advance, and that $d = 3$ then failed for $k = 7, 8$ while $d = 4$ succeeded, is the strongest evidence we have that the rank law is describing something real rather than fitting the cases it was measured on. The certificates are period-three matrices found by the construction above and verified on the full $2n \times 2n$ matrix, independently of the Fourier blocks used to find them. For $n = 60$ with roots at $\ell \in \{2, 3, 4, 5, 6, 8, 9\}$ the 120×120 matrix has its 14 smallest eigenvalues below 4×10^{-15} and its fifteenth at 5.1×10^{-2} — a separation of thirteen orders of magnitude — every vertex carries exactly three off-diagonal nonzeros, and the smallest required entry is 0.188 of the matrix scale. For $n = 72$ with $\ell \in \{1, 4, 6, 7, 9, 10, 11\}$ the corresponding figures are nullity 14, first nonzero eigenvalue 1.2×10^{-3} of scale, and smallest required entry 0.162. For $k = 7$ at $n = 96$, $d = 4$, $\ell \in \{3, 5, 6, 7, 8, 9, 10, 11\}$: nullity 16, sixteen eigenvalues below 1.4×10^{-14} , smallest required entry 0.107. For $k = 8$ at $n = 96$, $d = 4$, $\ell \in \{2, 3, 4, 5, 6, 7, 8, 9, 11\}$: nullity 18, first nonzero eigenvalue 1.25×10^{-2} of scale, smallest required entry 0.260. Several further root sets at each n give independent certificates — three at $k = 8$ alone, the best with smallest required entry 0.362 — so these are not isolated points but regions.

k	$2k + 2$	period d	$\lceil k/2 \rceil$	n	smallest required entry
6	14	3	3	60, 72	0.188, 0.162
7	16	4	4	96	0.107
8	18	4	4	96	0.260

Table 4: Period- d certificates attaining the ceiling for $k \geq 6$. The period used is in every case the one the rank law prescribes.

Why an exact certificate is hard here

The $k \geq 6$ certificates are numerical because their weights are algebraic numbers of no recognised form. The obvious remedy — find *rational* weights, so that the nullity can be confirmed by exact rational elimination as for $k = 4, 5$ — runs into a structural obstruction.

Proposition 41. *Let A be a period- d matrix on $P(n, k)$ with rational weights, and $m = n/d$. The Fourier blocks have entries in $\mathbb{Q}(\zeta_m)$, and the Galois automorphism $\sigma_j : \zeta_m \mapsto \zeta_m^j$ carries M_ℓ to $M_{j\ell}$. Since σ_j fixes the weights, $\sigma_j(\det M_\ell) = \det M_{j\ell}$, so*

$$\det M_\ell = 0 \implies \det M_{j\ell} = 0 \quad \text{for every } j \text{ coprime to } m.$$

Hence the set of singular blocks is a union of full Galois orbits of Γ_m , and a rational certificate can only realise a **Galois-closed** root set.

This is checked numerically in the form it can be: for rational weights the product of $\det M_\ell$ over a Galois orbit — the norm — comes out exactly rational, to machine zero, at $(k, d, n) = (6, 3, 60)$, $(6, 3, 72)$, $(6, 4, 64)$, $(5, 1, 24)$ and $(7, 4, 96)$.

The obstruction is now visible: exactness requires a Galois-closed root set, and Galois-closed root sets are precisely the ones that degenerate. All eight at $(k, d, n) = (6, 3, 72)$ hit the symbol to 10^{-15} with nullity 14 and all eight collapse a weight; both at $(6, 3, 60)$ likewise. So for these parameters *we found no rational certificate*. We state it that way deliberately: the degenerate points are singular points of the map and therefore attractors, which is why repeated random starts return to them, so a search collapsing every time is exactly the kind of negative evidence we caution against reading as a non-existence proof elsewhere in this section.

Relaxing to a subfield does not obviously help, and it is worth being precise about why. If the weights lie in $K = \mathbb{Q}(\zeta_m)^H$ then the singular set need only be H -stable, and at $(6, 3, 72)$ all three index-two subgroups admit realisable root sets. But that implication runs one way: an H -stable root set being realisable shows only that the necessary condition is not violated, and gives no reason to expect a certificate for it to have weights in K . Indeed integer-relation detection finds no such representation for the certificates we have, a search parametrised inside $\mathbb{Q}(\sqrt{2})$ from the start returns nothing, and at $(6, 3, 60)$ every index-two set degenerates outright. We therefore record exactness for $k \geq 6$ as **open**, with the evidence leaning negative, rather than as a route in progress. The remaining candidate is a computer-assisted existence proof (interval Newton) for an irrational certificate, which the ceiling makes tractable in principle, since certifying the equation $G(w) = G^*$ exactly delivers the nullity as a discrete consequence.

How many n does this reach?

The construction needs a divisor d of n with

$$\lceil k/2 \rceil \leq d \leq \frac{n}{2k+4},$$

the left inequality from the rank law and the right from requiring $\Gamma_{n/d}$ to contain $k + 1$ distinct interior values. This is a real restriction and it cuts both ways.

Proposition 42 (Reachability). *The set of n admitting such a divisor has density 1, but its complement contains every prime, and also every $n = 2p$ once $k \geq 6$ (since then $d \geq 3$ while the only small divisor is 2). Convergence is slow: at $k = 6$ the reachable proportion of n is 54% up to 240, 71% up to 1000 and 79% up to 5000; at $k = 8$, 36%, 62% and 73%.*

In particular **prime n lies outside the period- d method entirely**, because d must divide n and $d = 1$ gives rank 3. That is not a gap in the search but a structural feature: for prime n the only available construction is the fully non-equivariant one, i.e. solving $T = I$ directly in the $5n$ weights, which is the regime where our searches plateau. Since $Z(P(23, 5)) = 12 = 2k + 2$ with 23 prime, such n certainly can attain the ceiling; we simply cannot certify them this way.

Larger d is not merely permitted but helpful. The rank law requires only $d \geq \lceil k/2 \rceil$, and taking d strictly larger enlarges the solution manifold from dimension $5d - (k + 1)$, giving more room to avoid the degenerate locus. The effect is visible at the tightest configurations. At $k = 6$, $d = 3$, $n = 48$ the grid Γ_{16} offers exactly 7 interior values for the 7 required — a unique root set, no slack — and every solution collapses a spoke. At $k = 6$, $d = 4$, $n = 64$ the grid is again Γ_{16} with the same unique root set, but now the manifold has dimension $20 - 7 = 13$ rather than $15 - 7 = 8$, and a realisable certificate is found on the first attempt (smallest required entry 0.086, gap 6.8×10^{-2}). So the degeneracy of the previous subsection is a symptom of working at the minimum period, not an obstruction inherent to tight root sets. Certificates were likewise found at $k = 6$, $d = 6$, $n = 96$.

Within the reachable set, the construction appears to work essentially always. Sweeping n for $k = 6$ at $d = 3$ produced a realisable certificate at *every* n tested:

$$n = 60, 72, 84, 90, 96, 108, 120,$$

most of them on the first root set tried, with smallest required entries between 0.12 and 0.46. So

$$Z(P(n, 6)) = M(P(n, 6)) = 14 \quad \text{for each of these } n,$$

and the earlier appearance of a scattered list of congruence classes was an artefact of the cyclotomic route, not of the problem. We state this as a computation rather than a theorem: we have no proof that a realisable root set exists for every reachable n , only that one was found in every case examined.

Remark 43 (Status of Theorem 40). Unlike the cyclotomic certificates of Section 8, which are integer matrices whose nullity is confirmed by exact rational elimination, these are found by a Newton solve and their weights are algebraic numbers of no recognised form. We therefore record Theorem 40 as **numerically certified but not yet verified in exact arithmetic**. The evidence is strong — a thirteen-order spectral separation, the correct zero pattern with every required entry a fifth of the matrix scale, and the nullity never exceeding the ceiling $2k + 2$, which is an automatic check on the whole pipeline — but it is not of the same kind as the $k = 4$ and $k = 5$ results, and we do not claim that it is. The natural route to closing the gap is to choose the root set to be a union of full Galois orbits of Γ_m , which makes G^* a product of the Ψ_e and hence puts the system $G(w) = G^*$ over $\mathbb{Q}$; that is in progress.

11 Beyond congruence classes: a certified value at a prime

Theorem 38 says the cyclotomic layer cannot produce a value valid for all large n , and points at the replacement: Corollary 12 characterises the ceiling for *every* matrix on the pattern by the

condition $T = I$, which carries no arithmetic. This section carries that out at a prime n , where by Theorem 38(i) no period-one certificate can reach.

A better system to certify

Optimising $T = I$ is convenient but bad to certify: the map $w \mapsto T$ has a 51-dimensional image inside the 64-dimensional space of 8×8 matrices at $k = 3$, so $\{T = I\}$ is cut out by 13 local equations we cannot identify and no square subsystem is available. We certify the nullity directly instead.

Now $\text{null } A \geq r$ if and only if some full-rank $K \in \mathbb{R}^{2n \times r}$ has $AK = 0$. Fixing a row permutation P and writing K in graph form $K = P \begin{pmatrix} I_r \\ X \end{pmatrix}$ makes $\text{rank } K = r$ automatic, and

$$f(w, X) := A(w) K(X) = 0$$

is *bilinear*: quadratic of total degree 2, with rational coefficients. Its Jacobian is affine, so the Lipschitz constant of J is an explicit integer, and a Krawczyk contraction bound needs only three norms rather than an interval matrix product.

The choice of which equations to solve is not free. The system $AK = 0$ has $2nr$ equations of rank $2nr - \binom{r}{2}$, and the deficiency is structural: $K^\top AK$ is symmetric because A is, and $\binom{r}{2}$ is exactly its antisymmetric part. Concretely, if the $(2n - r)r$ equations on the non-pivot rows $P[r:]$ vanish then $K^\top AK$ equals the $r \times r$ block $(AK)[P[r:]]$, which must therefore be symmetric, so setting its upper triangle to zero forces the lower triangle as well. Hence

$$\underbrace{(AK)[P[r+m], t] = 0}_{(2n-r)r \text{ equations}} \quad \text{and} \quad \underbrace{(AK)[P[s], t] = 0, s \leq t}_{\binom{r+1}{2} \text{ equations}}$$

is *exactly equivalent* to $AK = 0$. Selecting rows by pivoting instead would leave $\binom{r}{2}$ equations unproved.

The reduction is not special to $P(n, k)$

Theorem 11 is stated for $P(n, k)$, but its proof uses only one structural fact: the row at an outer vertex meets *exactly one* inner variable, whose coefficient is a spoke weight and so is nonzero. Nothing in it requires the outer cycle to have voltage 1. Let $B_{p,q,w}$ denote the two-vertex base with a loop of voltage p at u , a loop of voltage q at v , and a single edge uv of voltage w ; its cover is cubic on $2n$ vertices, and $P(n, k) = B_{1,k,0}^n$.

Theorem 44 (Reduction on every two-vertex cover). *Let $n > 2(p + q)$ and let $A \in \mathcal{S}(B_{p,q,w}^n)$. Eliminating the inner coordinates leaves a single linear recurrence in the outer coordinates supported on the nine offsets*

$$\{-(p+q), -q, -(q-p), -p, 0, p, q-p, q, p+q\},$$

whose two extreme coefficients are nonzero. Consequently

$$\text{null } A = \dim \ker(T - I) \leq 2(p + q),$$

which is the degree span of $\det M(\zeta)$, with equality if and only if $T = I$. Theorem 11 is the case $p = 1, w = 0$, where $2(p + q) = 2k + 2$.

Proof. The neighbours of u_i are $u_{i\pm p}$ and v_{i+w} , so the row at u_i is $b'_{i-p}x_{i-p} + a_i x_i + b_i x_{i+p} + c_i y_{i+w} = 0$ with $c_i \neq 0$. Writing $i = j - w$ solves y_j as a three-term expression in x at offsets $-w - p, -w, -w + p$ from j . Substituting into the row at v_j , namely $e'_{j-q}y_{j-q} + d_j y_j + e_j y_{j+q} + c'_{j-w}x_{j-w} = 0$, and reindexing by $m = j - w$ gives exactly the nine listed offsets. The coefficient at $+(p + q)$ is $-e_j b_{j+q-w}/c_{j+q-w}$, a product of an inner edge weight, an outer edge weight and a spoke weight, all nonzero; symmetrically at $-(p + q)$. So the relation is a recurrence of order exactly $2(p + q)$, solvable in both directions, its solutions on $\mathbb{Z}$ form a space of that dimension on which T acts, and the n -periodic ones are its fixed vectors. If $p = q$ the nine offsets collapse to five, but the extreme coefficients are unchanged and so is the order. $\square$

The elimination was verified symbolically for (p, q, w) with $p \leq 3$, $q \leq 5$ and $w \leq 2$ — the offsets and the extreme coefficients come out as stated in every case — and the resulting bijection $x \mapsto (x, Lx)$ onto $\ker A$ was confirmed numerically, with $\text{null } A = \text{null } R$ and a lift residual at machine precision. See `verification/general_reduction.py`.

The hypothesis that each outer row meets exactly one inner variable is not, however, the real requirement. It makes the elimination *single-step*; what the argument actually needs is that the system can be advanced one fibre at a time. On a base with several u - v edges the elimination takes two steps instead of one, and still closes.

Theorem 45 (Reduction on theta covers). *Let B have two vertices joined by edges of voltages $t_1 < \dots < t_m$ and no loops, put $L = t_m - t_1$, and let $n > 2L$. For every $A \in \mathcal{S}(B^n)$ the kernel equations form a first-order recurrence on the state*

$$s_i = (x_i, \dots, x_{i+L-1}, y_i, \dots, y_{i+L-1}) \in \mathbb{R}^{2L},$$

so that $\text{null } A = \dim \ker(T - I) \leq 2L$, which is again the degree span of $\det M(\zeta)$.

Proof. The row at u_i is $a_i x_i + \sum_r w_i^{(r)} y_{i+t_r} = 0$; its highest-index inner variable is y_{i+t_m} , whose coefficient $w_i^{(r=m)}$ is an edge weight and so nonzero, and solving for it expresses y_{i+t_m} through x_i and the earlier y 's. The row at v_j is $d_j y_j + \sum_r w_{j-t_r}^{(r)} x_{j-t_r} = 0$; its highest-index outer variable is x_{j-t_1} , again with a nonzero coefficient. Shifting t_1 to 0, the two together carry s_i to s_{i+1} , so T_i exists and is invertible, and the n -periodic solutions are the fixed vectors of $T = T_{n-1} \cdots T_0$. Finally, with $f(\zeta) = \sum_r w^{(r)} \zeta^{t_r}$,

$$\det M(\zeta) = ad - f(\zeta)f(\zeta^{-1}),$$

whose degree span is $2(t_m - t_1) = 2L$. $\square$

Verified on eight bases, including four-edge ones, with $\text{null } A = \dim \ker(T - I)$ and the state dimension equal to the degree span in every case; see `verification/theta_reduction.py`. Together with Theorem 44 this settles the ceiling for two-vertex bases of both shapes — loops with a single connecting edge, and several connecting edges with no loops.

Theorems 44 and 45 are the two extreme shapes of a two-vertex base: loops with one connecting edge, and connecting edges with no loops. Between them lies the general case, where the two eliminations compete for the same variables, and there the answer is not uniform. It can be settled exactly, by counting.

Theorem 46 (Exactly when the reduction reaches the ceiling). *Let B carry loops of voltages $p, q \geq 1$ at u and v and connecting edges of voltages $0 = t_1 < \dots < t_m$, and let N be the order of the kernel recurrence. Then*

$$N = p + q + \max(p, t_m) + \max(q, t_m),$$

while the degree span is $D = \max(2(p + q), 2t_m)$. Consequently

$$N = D \iff t_m \leq \min(p, q),$$

and when the loops are absent $N = 2t_m = D$ always.

Proof. In a window of W consecutive fibres there are $2W$ unknowns. A row is usable only when every index it touches lies inside the window: the row at u_i needs $i \pm p$ and $i + t_r$ inside, leaving $W - p - \max(p, t_m)$ usable rows, and the row at v_j needs $j \pm q$ and $j - t_r$ inside, leaving $W - q - \max(q, t_m)$. The usable rows are generically independent, so the solution space has dimension $2W$ minus their number, which is the stated N , independent of W . The equivalence is arithmetic: if $t_m \leq \min(p, q)$ both maxima are the loop voltages and $N = 2(p + q) = D$; if $t_m > \min(p, q)$ then N exceeds both $2(p + q)$ and $2t_m$. $\square$

Verified against the measured order on sixteen bases with no discrepancy, and the equivalence on fourteen; see `verification/band_order.py`.

Remark 47 (What this does and does not decide). When $t_m \leq \min(p, q)$ the reduction proves null $A \leq D$ for every matrix carrying the pattern, and Theorems 44 and 45 are the cases $t_m = 0$ and “loops absent”. When $t_m > \min(p, q)$ the reduction still applies but yields only null $A \leq N$ with $N > D$. This does *not* exhibit a cover whose nullity exceeds the degree span: it shows the technique stops being tight there, not that the ceiling fails. Whether null $A \leq D$ still holds in that range is open, and is now a sharply posed question rather than a vague one — as is the case of three or more base vertices, which none of this touches.

Theorem 48 (Certified values off every congruence class). *For every n with $17 \leq n \leq 33$,*

$$Z(P(n, 3)) = M(P(n, 3)) = 8.$$

In particular this holds at the primes $n = 17, 19, 23, 29, 31$, and the range is consecutive: seventeen values with no exceptions.

Proof. $Z(P(17, 3)) \leq 8$ is the rotation bootstrap. For the lower bound, take $n = 17$, $k = 3$, $r = 8$, so $N = 34$ and the system above has $85 + 208 = 293$ unknowns and 272 equations, of which $208 + 36 = 244$ are the structural ones. At a numerically located point the Jacobian of the 244 structural equations has rank exactly 244, so fixing the remaining 49 unknowns at explicit rationals with denominator 10^4 leaves a square regular system. Newton in 60-digit arithmetic drives its residual to 3.2×10^{-61} ; the 28 dependent equations, which are not solved for, fall to 4.1×10^{-61} , confirming the structural reduction. With Y an approximate inverse of the square Jacobian,

$$\|Y\|_\infty = 1.54 \times 10^3, \quad \|I - YJ\|_\infty = 2.18 \times 10^{-12}, \quad \|Yf\|_\infty = 2.85 \times 10^{-60},$$

and with $L = 12$ bounding the Lipschitz constant of J in the ∞ -norm, the Krawczyk test on the box of radius $\delta = 10^{-12}$ gives $\alpha = \|I - YJ\|_\infty + \|Y\|_\infty L\delta = 1.85 \times 10^{-8} < 1$ and $\|Yf\|_\infty + \alpha\delta \leq \delta$. So a true zero (w^*, X^*) exists in that box. Then $A(w^*)K(X^*) = 0$ with rank $K = 8$, so null $A(w^*) \geq 8$, and ≤ 8 by Corollary 12. For that last step $A(w^*)$ must be a matrix of $P(17, 3)$, i.e. every edge weight nonzero, and this has to be known for w^* rather than for the box centre w_0 , which is all a direct computation gives. It follows from the box: $\|w^* - w_0\|_\infty \leq \delta$, so every edge weight of $A(w^*)$ is at least $\min_e |w_0(e)| - \delta$, and here $\min_e |w_0(e)|$ exceeds 0.05 times the matrix scale while $\delta = 10^{-12}$, so the bound survives with room to spare. The ninth singular value exceeds 0.126 times the scale, so the box is far from any degenerate stratum and the pattern is exactly that of $P(17, 3)$.

The remaining n are identical in structure and each passes the same test; for $n = 19$, to give one more set of numbers, there are 335 unknowns and 304 equations of which 276 are structural and 28 dependent, 59 unknowns are fixed at rationals, and $\|Y\|_\infty = 4.37 \times 10^2$, $\|I - YJ\|_\infty = 1.06 \times 10^{-12}$, $\|Yf\|_\infty = 1.20 \times 10^{-60}$, with Krawczyk passing at $\delta = 10^{-13}$ and $\alpha = 5.25 \times 10^{-10}$. See `verification/krawczyk.py` and `verification/certify_pipeline.py`. $\square$

Together with Theorem 24 this gives $Z(P(n, 3)) = 8$ at $n = 10$, every $17 \leq n \leq 33$, and every multiple of 10 — a set that no single divisibility condition describes. The search fails only at $n = 13$ and $n = 15$, where the solution variety has dimension 3 and 9 by (3), against 15 at $n = 17$ where it first succeeds. By Lemma 50 the slice searched is complete, so those are failures of the search and not evidence that no matrix of nullity 8 exists there. That distinction is not academic: $n = 18$ resisted the restart budget we used for its neighbours and certified once the budget was raised, so a failure here carries no information beyond the effort spent.

Remark 49 (Three independent numerical routes, one certificate). The certificate at $n = 17$ was located three separate ways — by minimising $\|T - I\|$ over all $5n$ weights, by an alternating solve of $AK = 0$ with the spokes normalised, and by the fully gauge-fixed search of (3) — and all three lead to a Krawczyk-certified matrix of nullity 8, with the gauge-fixed route giving the best conditioning ($\|Y\|_\infty = 2.1 \times 10^2$, ninth singular value 0.167 of the matrix scale, minimum edge weight 0.146). Since stage 1 is only a search and stages 2–3 are a proof, the numerical route taken does not enter the result; that three unrelated ones agree is a check on the search, not on the certificate.

Since 17 and 19 are prime, Theorem 38(i) shows no period-one certificate — in particular nothing in Sections 8 or 10 — can give these values. Indeed no divisibility class containing either of them is available at all: a class $\{n : L \mid n\}$ with $L \geq 3$ contains a prime only when L is that prime, and Γ_{17} offers only Ψ_1 and Ψ_{17} , the latter of degree $8 \neq k + 1 = 4$. These are the first values in this paper that do not sit on a divisibility class.

The reachable set stabilises

That $n = 17$ works and $n = 13$ does not is explained by a measurement rather than by arithmetic. The dimension of the reachable set of monodromies, that is the rank of $\partial T / \partial w$, grows with n and then *saturates*:

$k = 2, n$	5	6	7	8	9	10	11	12	≥ 13
rank	19	23	26	28	30	31	31	31	31
$k = 3, n$	7	9	11	13	14	15	16	17	≥ 18
rank	27	34	40	46	49	51	51	51	51

Past $n = 11$ for $k = 2$ and $n = 15$ for $k = 3$, increasing n adds no new reachable monodromies. So whether I is reachable stops depending on n in any essential way once n passes roughly $4k + 3$, and the observed pattern of successes and failures follows it: for $k = 3$ the ceiling is attained at $n = 17, 20$ and not at $n = 13 < 15$. Independently, at $n = 17$, $k = 3$ the solution variety of $f = 0$ has dimension $293 - 244 = 49 = 5n - \binom{2k+3}{2}$, exactly the codimension count for symmetric matrices on the pattern with nullity $2k + 2$.

There is also a hard *necessary* condition, from spending the gauge, and it pays to be careful about what the gauge can and cannot do.

Lemma 50 (A complete slice). *Every symmetric matrix on the pattern of $P(n, k)$ with all weights nonzero is carried, by a conjugation preserving symmetry, the pattern and the nullity, to one with*

every spoke equal to $+1$ and every outer weight equal to ± 1 — for even n , with a single outer weight left arbitrary. The signs may be taken to be $+1$ except possibly one, so for odd n there are two such slices and for even n one.

Proof. Two conjugations are available. Positive diagonal $A \mapsto DAD$ scales $b_i \mapsto d_i d_{i+1} b_i$, $c_i \mapsto d_i d_{n+i} c_i$, and sign diagonal $A \mapsto SAS$ with $S = \text{diag}(\pm 1)$ sends $b_i \mapsto s_i s_{i+1} b_i$, $c_i \mapsto s_i s_{n+i} c_i$. Since $d_i d_{n+i} > 0$, each spoke's *sign* is invariant under the first and must be fixed by the second: taking $s_{n+i} = s_i \text{sign}(c_i)$ makes every spoke positive, and choosing the inner d_{n+i} makes every spoke 1. The outer d_i then satisfy $u_i + u_{i+1} = -\log |b_i|$ with $u_i = \log d_i$, a cyclic system whose matrix is invertible for odd n and has corank 1 for even n ; in the latter case drop one equation, leaving one outer weight unnormalised. Finally $s_i s_{i+1} \text{sign}(b_i) = 1$ is solvable exactly when $\prod_i \text{sign}(b_i) = 1$, giving the two sign classes for odd n , while for even n the unnormalised weight absorbs the invariant. $\square$

We verified Lemma 50 numerically before relying on it: random matrices on the pattern are carried into the slice with the spokes exactly $+1$ and $|b_i| = 1$ to 10^{-15} , and the nullity unchanged, for both parities of n (`verification/gauge_fixed.py`). The point of the lemma is that a search restricted to the slice is not restricted at all, so a failure inside it is a failure of the search and not evidence that no matrix exists.

What remains after the fixing is $2n$ diagonal entries and n inner weights, with no residual gauge, and the solution variety of $f = 0$ has dimension

$$[3n + (2n - r)r] - [(2n - r)r + \binom{r+1}{2}] = 3n - (k + 1)(2k + 3),$$

which is nonnegative only when

$$n \geq \frac{(k + 1)(2k + 3)}{3} = 7, 12, 19, 26, 35, 46, 57 \quad (k = 2, \dots, 8). \quad (3)$$

Conjecture 51. *For every $k \geq 2$ there is $N(k)$ with $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for all $n \geq N(k)$.*

For $k = 2$ this is Theorem 20, with $N(2) = 11$, proved. We do not conjecture that $N(k)$ is linear: (3) forces $N(k) \geq \lceil (k + 1)(2k + 3)/3 \rceil$, which is quadratic, and at $k = 6$ that bound is 35 — larger than the value $4k + 3 = 27$ that the saturation data alone might suggest. The two quantities measure different things, and the dimension count is the binding one. Our data is consistent with $N(k)$ exceeding (3) by a small additive amount: $N(2) = 11$ against a bound of 7, and $14 \leq N(3) \leq 17$ against a bound of 12, since the ceiling is attained at $n = 17$ and 20 and not at $n = 13$.

The content of the conjecture for $k \geq 3$ is a reachability statement about products of transfer matrices, with no arithmetic in it — which is why we expect it to behave uniformly in n , unlike everything in Sections 8 and 10.

Remark 52 (A strict gap, and a correction to our own evidence). Two caveats. First, $n = 10$, $k = 2$ resists the search even though its reachable dimension is already saturated, and $Z(P(10, 2)) = 6$; so we expect $M(P(10, 2)) < Z(P(10, 2))$, a strict gap, though we have not proved it. Second, the eigenvalue-minimising search we reported earlier is unreliable and its negative results should not be read as evidence: it gave maximum nullity 7 at $P(20, 3)$, whereas the ceiling 8 is attained there both by Theorem 24 and, in the *symmetric* class, by the method of this section. It was collapsing onto the degenerate strata — all spokes zero, or all edges zero — that the formulations here exclude by construction.

A route to all large n

Conjecture 51 asks for one construction covering every large n . Certifying n one at a time cannot do that. But the condition $T = I$ is multiplicative along the cycle, and that suggests assembling the weight sequence from pieces each of which contributes the identity. The obstacle is that T_i is built from the weights in a window of width about $2k + 2$ around position i , so pieces interact across their boundaries and the product does not factorise. The following removes the obstacle.

Definition 53. Fix a *docking pattern*: values of (b, c, e, a, d) at $k + 1$ consecutive positions. A *tile* of length $\ell \geq 2(k + 1)$ is a weight sequence of length ℓ whose first $k + 1$ and last $k + 1$ positions carry the docking pattern. Its *interior* is the remaining $\ell - 2(k + 1)$ positions.

Lemma 54 (Tile independence). *Let the weight sequence on $\mathbb{Z}_n$ be a concatenation of tiles sharing one docking pattern. Then for each tile the product of the transfer matrices over its positions depends only on that tile's interior weights, and the monodromy is the product of these tile products.*

Proof. By Theorem 11, T_i is determined by the coefficients of the i th reduced equation, which involve b at indices in $[i - k - 1, i + k]$, c and a at $i - k, i, i + k$, e at $i - k$ and i , and d at i . So T_i depends only on weights at indices in $[i - k - 1, i + k]$. If the tile occupies $[s, s + \ell)$ and $i \in [s, s + \ell)$, this window is contained in $[s - k - 1, s + \ell + k - 1]$. The part below s lies in the preceding tile's final $k + 1$ positions and the part at or above $s + \ell$ lies in the following tile's first $k + 1$ positions; both carry the docking pattern, as do $[s, s + k]$ and $[s + \ell - k - 1, s + \ell)$. Hence every weight T_i sees is either docking or in this tile's interior. The second statement is associativity of the product $T = T_{n-1} \cdots T_0$ regrouped along the tiles. $\square$

Theorem 55 (Tiling). *Fix k and $L \geq 2(k + 1)$. Suppose that for every ℓ with $L \leq \ell < 2L$ there are interior weights, all of the required entries nonzero, making the tile product of length ℓ equal to the identity. Then*

$$Z(P(n, k)) = M(P(n, k)) = 2k + 2 \quad \text{for every } n \geq L,$$

with no congruence condition.

Proof. Every $n \geq L$ is a sum of integers from $[L, 2L)$: if $n < 2L$ take the single term n , and if $n \geq 2L$ take a term L and recurse on $n - L \geq L$. Concatenating the corresponding tiles gives a weight sequence on $\mathbb{Z}_n$ whose monodromy is, by Lemma 54, the product of the tile products, hence I . All required weights are nonzero, so the matrix carries the pattern of $P(n, k)$ and Theorem 11 applies; by Corollary 12, $\text{null } A = 2k + 2$. With $M \leq Z$ and the rotation bootstrap $Z \leq 2k + 2$, both are equal to $2k + 2$. $\square$

Lemma 54 was checked before use: perturbing one tile's interior leaves the other tiles' products unchanged to the last bit, 0 exactly, for $k = 2, 3, 4$ (`verification/tiles.py`). Two remarks on the shape of Theorem 55. First, the hypothesis over a full interval $[L, 2L)$ is what makes the conclusion gapless; using two tile lengths a, b with $\gcd(a, b) = 1$ instead would give only n of the form $ia + jb$, leaving every $n < (a - 1)(b - 1)$ uncovered, which for the smallest solvable lengths at $k = 3$ is already several hundred. Second, the hypothesis is a *finite* list of independent finite problems: a tile of length ℓ has $5(\ell - 2(k + 1))$ free weights and the requirement is $(2k + 2)^2$ equations, so it is solvable on parameter count once ℓ is a little above $2(k + 1) + (2k + 2)^2/5$. For $k = 3$ that is $\ell \geq 21$, and $L = 21$ gives every $n \geq 21$.

The tiles exist, and are certified

We carried the computation out for $k = 3$. The parameter count is sharp in practice: at $\ell = 19, 20$ the systems are underdetermined (55 and 60 unknowns against 64 equations) and the residual stalls near 4×10^{-1} ; and from $\ell = 21$, the first length with more unknowns than equations, identity tiles are found. A Newton solve produced one for *every* length in the required range, $21 \leq \ell \leq 41$, which is exactly $[L, 2L]$ for $L = 21$; we in fact have tiles for every $21 \leq \ell \leq 43$.

Each of those 23 tiles is then *certified*, which needs one observation to avoid interval arithmetic on a rational map. The condition $T_{\text{tile}} = I$ is a rational equation in the tile weights, so a Krawczyk bound on it directly would require interval matrix products. It is unnecessary: by Corollary 12 the monodromy of the cyclic matrix built from a *single* tile on $P(\ell, k)$ is the tile product, so

$$T_{\text{tile}} = I \iff \text{null } A_{\text{one tile}} = 2k + 2,$$

and the right-hand side is certified by the bilinear machinery of Theorem 48, whose Jacobian is affine. The only extra requirement is that the tiles still compose, which needs them all to carry the same docking pattern; this is arranged by pinning the $5(2k + 2)$ docking weights at their exact integer values ($b = c = e = 1$, $a = d = 0$) rather than letting the solver move them, so they belong to the fixed set and the certified solution carries them exactly.

All 23 certifications at $k = 3$ pass. For $\ell = 22$: 398 unknowns, 352 equations of which 324 are structural and 28 dependent, 40 docking pins and 74 unknowns fixed at rationals in all; high-precision Newton drives the structural residual to 3.1×10^{-61} and the dependent equations to 2.7×10^{-61} ; $\|Y\|_{\infty} = 5.4 \times 10^2$ and the Krawczyk test passes with $\alpha = 6.7 \times 10^{-11}$. Across all lengths $\alpha \leq 2.2 \times 10^{-10}$, every certified matrix has nullity exactly 8, every docking weight agrees with its prescribed integer to 10^{-14} , and the minimum edge weight is 0.15 of the matrix scale. See `verification/certify_tiles.py`.

Remark 56 (The tile equations are degenerate, and by how much). The system $T_{\text{tile}} = I$ has $(2k + 2)^2$ equations but they are not independent. Measuring the rank of the Jacobian of $w \mapsto T_{\text{tile}}$ at a solution gives exactly

$$\text{rank} = \binom{2k + 3}{2}, \quad \text{deficiency} = \binom{2k + 2}{2},$$

namely 36 and 28 at $k = 3$ and 55 and 45 at $k = 4$, at every tile length tested. Two consequences. First, it explains the behaviour of the solve: the residual has $\binom{2k+2}{2}$ components that vanish automatically near a solution, so a least-squares step is working on a degenerate system, and that is consistent with the erratic convergence reported in the remark below. Second, the count that a tile must satisfy is $5(\ell - 2(k + 1)) \geq \binom{2k+3}{2}$ rather than $\geq (2k + 2)^2$, which for $k = 3$ would permit $\ell \geq 16$ rather than $\ell \geq 22$ and so a smaller L . Whether short tiles exist is a separate question from whether the count allows them, and the answer is in between. With a larger restart budget $\ell = 21$ *does* yield an identity tile, certified like the rest, so $L = 21$ rather than 22 at $k = 3$ and the required range is $[21, 42]$; this does not change Theorem 57, which already reaches $n \geq 17$ through Theorem 48. Below that the searches fail and the failures look different in kind: $\ell = 19$ and 20 stall at 1.4×10^{-1} and 3.7×10^{-1} , and $\ell = 16, 17, 18$ at 8.3×10^{-1} , 7.8×10^{-1} and 8.2×10^{-1} , whereas every length that we eventually solved first appeared as a residual near 10^{-2} or below. So the count is not the binding constraint and I appears not to be a product of fewer than about twenty transfer matrices. We record that as a computation and not a proof: we have no lower bound on the number of factors needed to write I , and $\ell = 21$ is a reminder that a stalled search at these sizes can mean only that the budget was too small.

We note that $\binom{2k+3}{2} = \dim Sp(2k+2)$, and that the space of skew forms B with $T^\top BT = B$ has dimension $k+1$. It is tempting to conclude that T is symplectic with respect to a weight-dependent form. We do not claim it: our attempts to exhibit a nondegenerate invariant form were numerically unstable, giving residuals of 10^{-12} at $k = 2, 4, 5$ but 10^{-1} at $k = 3$, so the coincidence of dimensions is suggestive and no more.

Theorem 57 (All large n , for $k = 3$ and $k = 4$).

$$Z(P(n, 3)) = M(P(n, 3)) = 8 \quad \text{for every } n \geq 17,$$

$$Z(P(n, 4)) = M(P(n, 4)) = 10 \quad \text{for every } n \geq 29.$$

Proof. For $k = 3$, identity tiles exist for every ℓ with $21 \leq \ell < 42$, and for $k = 4$ for every ℓ with $29 \leq \ell < 58$; each is certified as above. (We in fact have certified tiles for every $21 \leq \ell \leq 43$ at $k = 3$ and every $29 \leq \ell \leq 61$ at $k = 4$, comfortably more than $[L, 2L)$ requires.) A certification exhibits a box containing a true real solution of $AK = 0$ with K of full rank $2k+2$, so the one-tile cyclic matrix has nullity at least $2k+2$ and hence exactly $2k+2$ by Corollary 12, so $T_{\text{tile}} = I$; and its docking weights are the prescribed integers exactly, since they are held fixed throughout, so tiles with different ℓ compose. Theorem 55 with $L = 21$ and $L = 29$ respectively gives $n \geq 21$ and $n \geq 29$, and Theorem 48 covers $17 \leq n \leq 20$ at $k = 3$. $\square$

All $23 + 33 = 56$ certifications pass. To give the numbers at both ends: at $k = 3$, $\ell = 22$ has 398 unknowns and 352 equations of which 324 are structural and $28 = \binom{8}{2}$ dependent, with 40 docking pins, high-precision residual 3.1×10^{-61} , and Krawczyk $\alpha = 6.7 \times 10^{-11}$; at $k = 4$, $\ell = 61$ has 1425 unknowns and 1220 equations of which 1175 are structural and $45 = \binom{10}{2}$ dependent, with 50 docking pins, residual 4.7×10^{-61} , and $\alpha = 3.7 \times 10^{-10}$. Across all 56, $\alpha \leq 1.0 \times 10^{-9}$, every certified matrix has nullity exactly $2k+2$, every docking weight matches its prescribed integer to 10^{-14} , and the minimum edge weight is 0.15 of the matrix scale. See `verification/certify_tiles.py`.

These are the statements Conjecture 51 asks for, and they are proofs rather than computations reported as such: the chain is Theorem 55, Lemma 54 and Corollary 12, all proved, together with 56 certifications whose Krawczyk hypotheses are verified in exact rational arithmetic (Section 11). No step rests on a floating-point nullity. With Theorem 20 for $k = 2$, the three smallest k beyond the trivial case are now settled for all large n , with

$$L(2) = 9, \quad L(3) = 17, \quad L(4) = 29,$$

against the tile lower bound $2(k+1) + (2k+2)^2/5$, which is 13.2, 20.8 and 30.

Theorem 58 (Complete determination for $k = 2$, $k = 3$ and $k = 4$). *For every n with $2k < n$, the value $Z(P(n, k))$ is now known when $k = 2$, $k = 3$ or $k = 4$:*

n	5	6	7	8	9	10	11	≥ 12	n	7	8	9	10	11	12	≥ 13
$Z(P(n, 2))$	5	4	6	5	6	6	6	6	$Z(P(n, 3))$	6	6	6	8	7	7	8
n	9	10	11	12	13	14	15	16	17	≥ 18						
$Z(P(n, 4))$	6	6	7	6	8	8	9	8	9	10						

No value is left open in any of the three rows.

Proof. The tails are Theorem 20 for $k = 2$, Theorem 59 for $k = 3$, and Theorem 57 for $k = 4$, which gives $Z(P(n, 4)) = 10$ for every $n \geq 29$; the finitely many remaining values, including $18 \leq n \leq 28$ at $k = 4$, are exhaustive searches over vertex subsets. $\square$

For $k = 2$ this completes what [6] began, which gave $Z(P(n, 2)) = 6$ only for $n \geq 10$. For $k = 3$ it is the first determination of any kind: the value claimed in [6] for that case is false, and [7], which found the error, leaves the threshold open. Note that neither row is monotone and neither is eventually constant from the first time the bound is attained — $k = 3$ reaches 8 at $n = 10$, drops to 7 at $n = 11, 12$, and only then settles. Any argument that assumed monotonicity would have got this wrong, and one did.

The same exhaustive computation at $k = 4$ gives

n	9	10	11	12	13	14	15	16	17	18
$Z(P(n, 4))$	6	6	7	6	8	8	9	8	9	10

n	19	20	21	22	23	24	25	26	27	28
$Z(P(n, 4))$	10	10	10	10	10	10	10	10	10	10

which is more erratic still: the value falls at $n = 12$ and again at $n = 16$, and the bound 10 is first attained at $n = 18$. These reach $n = 28$ because the rotation $\rho(i) = i + 1$ lets the search assume $u_0 \in S$, or $v_0 \in S$ if S is entirely inner, which is exhaustive up to symmetry and so still exact; at $n = 28$, ruling out every 9-set takes about 1.2×10^9 candidates. Together with Theorem 57 for $n \geq 29$ this leaves *no* value of $Z(P(n, 4))$ undetermined, which is why $k = 4$ appears in Theorem 58.

The matrix statement is weaker than the forcing statement at exactly one value. $M(P(n, 4)) = 10$ is proved for every $n \geq 29$ by Theorem 57, for $n = 25, 26, 27, 28$ by individual certificates, and for every n divisible by 60, 70 or 90 by Theorem 35. Only at $n = 24$ do we know $Z = 10$ without a certificate, so $M(P(24, 4))$ is not determined; nothing suggests $M < Z$ there, we simply have not found the matrix.

Theorem 59 ($k = 3$ completely, and the stabilization threshold).

$$Z(P(n, 3)) = 8 \quad \text{for every } n \geq 13,$$

and 13 is optimal: $Z(P(11, 3)) = Z(P(12, 3)) = 7$. Moreover $Z(P(n, 3)) = M(P(n, 3)) = 8$ for every $n \geq 17$ and for every n divisible by 10.

Proof. For $n \geq 17$ this is Theorem 57. For $n = 13, 14, 15, 16$ exhaustive search over vertex subsets shows no forcing set of size 7 exists, so $Z = 8$ by the bound $Z \leq 2k + 2$. The values $Z(P(11, 3)) = Z(P(12, 3)) = 7$ are likewise exhaustive. $\square$

This proves Conjecture 5 of [7], stated there as

“Conjecture 5. $Z(P(n, 3)) = 8$ for every $n \geq 13$,”

and described in that note’s abstract as needing “a lower-bound proof valid for all large n ”. Theorem 59 is that statement, and the missing ingredient is exactly what Section 11 supplies: $M \leq Z$ turns a matrix into a lower bound, and the tiling construction produces such matrices for all large n at once rather than one n at a time. The upper bound $Z \leq 8$ for $n \geq 9$ is already in [7], as are the values for $7 \leq n \leq 20$ by exhaustive search; what is new here is the lower bound for every $n \geq 13$, and with it the stabilization threshold, which the same note records as open alongside a rigorous lower bound for $k \geq 4$. It is worth being exact about what that note does and does not claim, because it is unusually careful on the point. Of Conjecture 5 it writes that “the content is the lower bound $Z(P(n, 3)) \geq 8$ for all $n \geq 13$, which we verified only for $n \leq 20$ ”, that proving it “means excluding 7-vertex forcing sets uniformly in n ”, and that “we will not assert it from finite computation, which would reproduce the gap this note corrects”. So the missing ingredient

is named precisely, and extrapolation from finite data is explicitly declined. Uniformity in n is exactly what tiling provides: one finite family of certified tiles yields a matrix for every $n \geq L$ at once. The sequence is genuinely non-monotone below it — $Z(P(n, 3))$ runs 6, 6, 6, 8, 7, 7, 8, 8, ... for $n = 7, \dots, 14$ — which is why the threshold is not simply the first n attaining the bound, and why Conjecture 8 is stated in the form it is.

The certification is exact, not floating-point

A Krawczyk bound is a theorem about real numbers, and evaluating its three norms in double precision is not a proof of its hypotheses. Our first implementation did exactly that: $\|Y\|_\infty$, $\|I - YJ\|_\infty$ and $\|Yf\|_\infty$ were computed in float64 with no directed rounding, and J and f were evaluated at the double-rounded point rather than at the high-precision one. The margins are enormous — α near 10^{-10} against a threshold of 1 — so the conclusion was surely true, but “certified” should not rest on a margin.

It need not. Three observations make the whole test exact at essentially no cost.

1. *The box centre is exactly rational.* A stored certificate is a float64 vector, and every float64 is a dyadic rational, so z_0 is known exactly. We snap it to the grid $2^{-60}\mathbb{Z}$, which moves z_0 by at most 2^{-61} and is harmless because z_0 is only the centre of the box and the test re-measures $f(z_0)$ after the move.
2. *f and J have integer coefficients.* Since $f(w, X) = A(w)K(X)$ is bilinear, J is affine and every entry of $J(z_0)$ is 0, 1, or a single coordinate of z_0 . So $J(z_0)$ and $f(z_0)$ are exactly rational with denominators 2^{60} and 2^{120} , and are built here as integer matrices.
3. *Y is arbitrary.* The Krawczyk test holds for *any* matrix Y ; Y has only to make $\|I - YJ\|_\infty$ small, and need not be $J(z_0)^{-1}$. So we take the float64 inverse and read it as an exact dyadic rational, losing nothing whatsoever.

Hence $\|Y\|_\infty$, $\|I - YJ(z_0)\|_\infty$ and $\|Yf(z_0)\|_\infty$ are exact rationals, and the two comparisons $\alpha < 1$ and $\|Yf\|_\infty + \alpha\delta \leq \delta$ are exact integer comparisons. The one obstacle is speed, since α needs every entry of the $\text{rk} \times \text{rk}$ product $YJ(z_0)$ with rk up to 1175. We multiply by *balanced limb decomposition*: each integer is written in base 2^{20} with digits in $[-2^{19}, 2^{19})$, each pair of digit matrices is multiplied in float64 — those products stay below 2^{53} and are therefore themselves exact integers — and the limb products are recombined with exact integer arithmetic. This gives the exact integer product at the speed of BLAS; the routine refuses to run if the limb bound is not met, so it cannot silently return an inexact answer.

All 56 tile certifications and all 17 per- n certifications of Theorem 48 pass the test evaluated this way, with $\alpha \leq 1.0 \times 10^{-9}$. To give the two that the thresholds rest on: $\ell = 21$ at $k = 3$ has $\text{rk} = 308$, $\|Y\|_\infty = 1.5743 \times 10^3$, $\alpha_0 = 2.5071 \times 10^{-11}$ and $\alpha = 2.139912 \times 10^{-10}$; $\ell = 29$ at $k = 4$ has $\text{rk} = 535$, $\|Y\|_\infty = 1.4416 \times 10^3$, $\alpha_0 = 9.2248 \times 10^{-12}$ and $\alpha = 1.822215 \times 10^{-10}$. These agree with the float64 values to three figures, which is the expected outcome and the reason the original conclusion was not in doubt; the point is that they are now *bounds* rather than estimates. See `verification/exact_krawczyk.py` and `verification/exact_certify.py`.

Two smaller points belong here. First, the Lipschitz constant. Row (i, t) of f involves the ≤ 4 weights whose cells meet row i — one diagonal and three edges, since $P(n, k)$ is cubic — each contributing one entry of K , and the ≤ 4 columns $X[m, \cdot]$ for which $A[i, \ell] \neq 0$, each contributing one weight. Both groups are Lipschitz 1 in the corresponding coordinate, so the row sum of $|J(z_1) - J(z_2)|$ is at most $8\|z_1 - z_2\|_\infty$ and $L = 8$ is valid; it is also attained, in the all-ones

direction. We nevertheless run every test with $L = 12$, which is conservative: a larger L makes α larger and the test strictly harder to pass, so using 12 can only understate the margin.

Second, the nonvanishing of the edge weights is deduced over the whole box exactly as in the proof of Theorem 48, and for the tiles the docking weights are not perturbed at all: they sit among the fixed coordinates, so they are exact integers at z_0 and the box moves only the free coordinates. That is what lets tiles of different lengths compose — the certified w^* carries the docking pattern exactly, not approximately.

What a general existence proof would need, and what it reduces to

Theorems 55 and 57 together have an unsatisfying shape: the implication is proved for all k , but its hypothesis is verified by *computing* the tiles, separately for each k . A proof that identity tiles exist for every k and every sufficiently long ℓ would give $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for all k and all $n \geq L(k)$ in one statement, and would subsume Sections 8 and 10 entirely. This subsection identifies what that proof is a proof *about*.

Fix k and a docking pattern, and for $\ell \geq 2(k + 1)$ let $\mathcal{S}_\ell \subseteq \text{GL}_{2k+2}(\mathbb{R})$ be the set of tile products of length ℓ realisable with all weights nonzero. Existence of an identity tile of length ℓ is exactly the statement $I \in \mathcal{S}_\ell$. Three facts constrain $\mathcal{S}_\ell$.

Observation 60 (Symplectic structure). For a fixed docking pattern there is a skew-symmetric B , unique up to scale, with

$$P^\top B P = B \quad \text{for every } P \in \mathcal{S}_\ell \text{ and every } \ell,$$

and B is nondegenerate. Hence $\mathcal{S}_\ell \subseteq \text{Sp}(B) \cong \text{Sp}(2k + 2, \mathbb{R})$, a connected simple Lie group of dimension $\binom{2k+3}{2}$.

The mechanism is not special to this problem: the recurrence of Theorem 11 is self-adjoint, because A is symmetric, and the transfer matrices of a self-adjoint difference system carry a Wronskian-type skew form from one step to the next. A tile whose two ends carry the same docking therefore returns that form to itself. Concretely, solving $T_i^\top B_{i+1} T_i = B_i$ along a tile has a $(k + 1)$ -dimensional solution space for any one tile, and imposing the same docking across many different interiors cuts it to the single B above.

We label this an observation rather than a theorem: we verified it numerically for $k = 2, \dots, 6$ at two tile lengths each, finding in every case exactly one invariant skew form, nondegenerate, preserved to between 10^{-13} and 10^{-8} , and we have not written out the algebraic proof (`verification/symplectic_structure.py`).

Observation 61 (Full dimension). The rank of the differential of $w \mapsto P_\ell(w)$ equals $\binom{2k+3}{2}$ exactly — the dimension of $\text{Sp}(B)$ — for every (k, ℓ) we computed, $2 \leq k \leq 6$. So for ℓ past the threshold of Remark 56, $\mathcal{S}_\ell$ has nonempty interior in $\text{Sp}(B)$.

Lemma 62 (Concatenation). $\mathcal{S}_m \cdot \mathcal{S}_\ell \subseteq \mathcal{S}_{\ell+m}$.

Proof. Immediate from Lemma 54: the docking pattern makes the product over each tile independent of its neighbours, so concatenating a tile of length m after one of length ℓ realises the product as a tile of length $\ell + m$. $\square$

Together these say that $\mathcal{S} = \bigcup_\ell \mathcal{S}_\ell$ is a subsemigroup of $\text{Sp}(2k + 2, \mathbb{R})$ with nonempty interior, and the question “do identity tiles exist?” becomes

is I in the interior-carrying subsemigroup generated by these transfer matrices?

That is a controllability question about a Lie semigroup, and the honest status is that it does not follow from the above alone: $Sp(2n, \mathbb{R})$ does admit proper subsemigroups with nonempty interior — compression semigroups of convex cones are the standard examples — so one must show the generators are not confined to such a wedge. What the observations do establish is that the group is identified, the dimension obstruction is absent, and the remaining question is of a recognised type rather than an unstructured search. We note that I is in fact reachable for every k where we have looked ($k = 3$ at all $21 \leq \ell \leq 43$, $k = 4$ at all $29 \leq \ell \leq 61$, $k = 5$ at 21 lengths in $[54, 75]$), which is evidence for the general statement and not a proof of it.

Remark 63 (What this does not give). Theorem 57 leaves $7 \leq n \leq 16$ open at $k = 3$, apart from $n = 10$ which Theorem 24 settles; the construction of Section 11 fails at $n = 13$ and 15, where (3) gives the solution variety dimension only 3 and 9. At $k = 4$ it leaves $9 \leq n \leq 30$ open apart from the multiples of 60, 70 and 90 covered by Theorem 35.

At $k = 5$ the picture is different from what we first reported, and the correction is worth stating plainly. The count $\ell \geq 2(k+1) + (2k+2)^2/5$ gives $\ell \geq 41$ at $k = 5$ and $\ell \geq 54$ at $k = 6$, and we initially searched only $41 \leq \ell \leq 45$ at $k = 5$ and found no identity tile, the residual stalling between 7×10^{-2} and 2×10^{-1} . We drew no conclusion from that, correctly: the count is a *lower bound* on where tiles can exist, not a prediction of where they start, and at $k = 3$ the gap is already large (21 against a count of 20.8, but nothing at all below 21). Searching higher, we now have identity tiles at $k = 5$ for

$$\ell = 54, 55, \dots, 72, 74, 75,$$

21 lengths in all, and *all of them are certified*, by the same pipeline and to the same standard as $k = 3$ and $k = 4$: the exact Krawczyk test of Section 11 passes at every one, with α up to 8.4×10^{-10} , nullity exactly 12, and the docking pattern exact.

This is enough for an all-large- n statement, and by a route worth spelling out because it shows the $[L, 2L)$ hypothesis of Theorem 55 is sufficient rather than necessary. We are missing $\ell = 73$ and $\ell = 76, \dots, 107$, so $[54, 108)$ is *not* covered and Theorem 55 does not apply as stated. But what the construction actually needs is only that n be a *sum* of certified lengths, and the numerical semigroup generated by $\{54, \dots, 72, 74, 75\}$ contains every integer $n \geq 162$. Hence:

Theorem 64 ($k = 5$, all large n).

$$Z(P(n, 5)) = M(P(n, 5)) = 12 \quad \text{for every } n \geq 162,$$

and also for every n in $[54, 162)$ that is a sum of certified tile lengths, which is 64 further values.

Proof. Identical to Theorem 57 except for the last step. Each certified tile gives a one-tile cyclic matrix of nullity exactly $2k + 2 = 12$ by Corollary 12, hence $T_{\text{tile}} = I$, and the docking weights are the prescribed integers exactly, so tiles of different lengths compose (Lemma 54). Concatenating tiles of lengths summing to n therefore gives a matrix on $P(n, 5)$ of nullity $2k + 2$. Instead of invoking $[L, 2L)$ we check the semigroup directly: the largest integer not representable as a sum of $\{54, \dots, 72, 74, 75\}$ is 161, so every $n \geq 162$ is representable. See `verification/verify_all.py`, section 9. $\square$

The threshold 162 is certainly not optimal — it is an artefact of the two missing lengths, and filling $\ell = 73$ and $76 \leq \ell \leq 107$ would bring it down to 54. We report it as it stands rather than waiting, since it is a theorem either way.

Nothing in Lemma 54 or Theorem 55 refers to k , and the certification scaled from 8×8 transfer matrices at $k = 3$ through 10×10 at $k = 4$ to 12×12 at $k = 5$ with no modification at all.

We considered the natural explanation — that the product of ℓ transfer matrices becomes too ill-conditioned for the residual $T - I$ to carry information — and it does not survive contact with the numbers: at $k = 5$, $\ell = 44$ the product has condition number about 2.5×10^6 , whereas at $k = 4$, $\ell = 61$, which *does* yield a certified tile, it is about 1.1×10^{10} . So conditioning is not the discriminating factor and we do not have a diagnosis. The most likely mundane explanation is restart budget: we gave each $k = 5$ length four to six restarts, and at $k = 4$ every length that failed at that budget succeeded when given fifteen to sixty. On the evidence we have, the right statement is that we did not find $k = 5$ tiles, not that they do not exist.

One practical observation, since it governs how much compute the method needs. The Newton solve for a tile is erratic: at the restart budget we first used, roughly half the lengths at $k = 4$ failed, including lengths with *more* unknowns than ones that succeeded. Every such failure that we retried with a larger budget — $\ell = 31, 32, 33, 35, 38$ — then solved to 10^{-14} . So a failed tile search at these sizes carries no information beyond the effort spent, and we were careful not to read the first pass as evidence about $k = 4$. Extending a neighbouring tile as a starting point does *not* substitute for restarts: the damped Newton stalls near 10^{-2} from such starts at the lengths that needed retrying.

12 Why lower bounds resist standard methods

A set is zero forcing iff it meets every *fort* — a nonempty F such that every vertex outside F has 0 or ≥ 2 neighbours in F — so Z is a minimum hitting set over forts. Forts, and the covering formulation of Z they give, are due to Brimkov, Fast and Hicks [4]. Three standard consequences all fail here, and all degrade as k grows.

- **Disjoint forts.** Pairwise disjoint forts $F_1, \dots, F_t$ force $Z \geq t$. But forts in $P(n, k)$ have 7–13 vertices while the graph has $2n \leq 56$, and a greedy search found at most *two* disjoint forts in every case tested: the certificate cannot exist at these sizes.
- **The fort LP.** Computing the exact LP value by cutting planes gives $\text{LP}/Z \approx 0.43\text{--}0.50$, e.g. $\text{LP} = 3.00$ against $Z = 7$ for $P(12, 3)$, and 3.21 against $Z = 10$ for $P(18, 4)$. The integrality gap is about 2 and widens with k . This also explains why fort-based exact solvers converge slowly on this family: branch and bound must close a factor-2 gap by branching.
- **Spectral.** The Hoffman-type bound $Z \geq N(1 + 2\lambda_{\min}/(d - \lambda_{\min}))$ yields values in 1.2–2.9 against true Z of 6–12, and also weakens as k grows.

13 Status of the claims

Every claim below is labelled by the standard it meets. Four standards appear, in decreasing strength: *proved*, meaning a proof in the text; *proved by an exact certificate*, meaning an explicit matrix whose nullity is computed in $\mathbb{Z}$ or $\mathbb{Q}(\zeta_n)$; *proved by interval certification*, meaning a Krawczyk contraction bound establishing that a true real solution exists in an explicit box; and *numerical*, meaning a floating-point computation only. We also list what we have refuted, including our own earlier claims.

- **Proved.** Theorems 3, 7 and Theorem 11 with Corollary 12; Propositions 13 and 14; Theorem 29; Theorem 30 and Theorem 26, the nullity classifications of the adjacency matrix and of its signed variant; Theorem 16, that $c^2 = 0$ if and only if the symbol factors as $(s + a)(L_k + t)$,

with Corollaries 17 and 18 and Proposition 19; Theorem 28, that no rational period-one symbol attains 14 at $k = 6$ in either matrix class; Theorem 38, that a period-one certificate covers exactly one divisibility class, which contains at most one prime, together with the identity $e_2 = -k$ for every $k \geq 3$; and Theorem 31 with Theorem 32 for cyclic covers.

- **Proved by an exact certificate.** $Z(P(n, 2)) = 6$ for every $n \geq 9$ with $n \neq 10$, and for every n divisible by 7 (Theorem 20); $Z(P(n, 3)) = 8$ for every n divisible by 10 (Theorem 24); $Z(P(n, 4)) = 10$ for every n divisible by 60, 70 or 90; $Z(P(n, 5)) = 12$ for every n divisible by 24; $Z(P(n, 7)) = 16$ for every n divisible by 120 (Theorem 27). The $k = 4, 5$ values were the first exact values for $k \geq 4$; the $k = 7$ value replaces a floating-point claim we previously made at a single n . In the $k = 2$ ($7 \mid n$), $k = 3$ and $k = 7$ cases the certificate is an integer matrix with entries in $\{0, \pm 1\}$.
- **Proved by exact certification.** $Z(P(n, 3)) = M(P(n, 3)) = 8$ for every $n \geq 17$, and $Z(P(n, 4)) = M(P(n, 4)) = 10$ for every $n \geq 29$ (Theorem 57), and $Z(P(n, 5)) = M(P(n, 5)) = 12$ for every $n \geq 162$ (Theorem 64), from Theorem 55 together with certified identity tiles of every length in $[21, 44)$ at $k = 3$ and $[29, 62)$ at $k = 4$, plus Theorem 48 for $17 \leq n \leq 20$; at $k = 5$ the certified lengths are $[54, 73) \cup \{74, 75\}$, which do not form an interval $[L, 2L)$ but generate every $n \geq 162$ as a sum. No congruence condition appears and the primes are included. All 77 tile certifications — 23 at $k = 3$, 33 at $k = 4$, 21 at $k = 5$ — and all 17 per- n certifications are Krawczyk contraction bounds with $\alpha \leq 1.0 \times 10^{-9}$, each evaluated in *exact rational arithmetic* (Section 11); no step rests on a floating-point nullity. Each rests on a Krawczyk contraction bound for the bilinear system $A(w)K(X) = 0$, with the square subsystem chosen structurally so that it is equivalent to $AK = 0$ rather than merely a maximal-rank selection of it. By Theorem 38 no period-one certificate can reach these n at all, so this is the only route to them that we know.
- **Proved (finite).** Every tabulated value of Z , by exhaustive search over vertex subsets. Combined with the above this gives Theorem 58: $Z(P(n, k))$ is determined for every admissible n when $k = 2$, $k = 3$ and $k = 4$, with no value left open in any of the three rows. In particular Theorem 59, $Z(P(n, 3)) = 8$ for every $n \geq 13$ with 13 optimal: the four values $n = 13, 14, 15, 16$ are exhaustive searches and the rest is Theorem 57, so the stabilization threshold at $k = 3$ — recorded as open in [7] — is exactly 13. This is the one result here whose statement needs no matrix theory to check.
- **Numerical, not exact.** Theorem 40: $Z(P(n, 6)) = 14$ at $n = 60, 72, 84, 90, 96, 108, 120$ and $Z(P(96, 8)) = 18$. The certificates are period-three matrices whose weights come from a Newton solve, verified on the full $2n \times 2n$ matrix with a wide spectral separation and every required entry a substantial fraction of the matrix scale, but not in exact arithmetic. We do not claim these at the standard of the items above. ($Z(P(96, 7)) = 16$ was in this category and is now exact.)
- **Observation supported by computation, not proved.** The rank law $\text{rank} = \min(2d + 1, k + 1)$, verified for $3 \leq k \leq 13$ and $1 \leq d \leq 7$. Also the saturation of the reachable set of monodromies, at dimension 31 for $k = 2$ and 51 for $k = 3$.
- **Conjectural.** Conjecture 8, and Conjecture 51 for $k \geq 6$. For $k = 3$ and $k = 4$ the conjecture is a theorem (Theorem 57), and for $k = 5$ it is a theorem with a non-optimal threshold (Theorem 64, $N(5) \leq 162$); what remains open is whether $N(k)$ exists for every k , for which we prove only the necessary condition (3).

- **Refuted (ours).** That the period-one construction “never exceeds 6 for every $k \geq 2$ ”. Three roots can be prescribed and no more, but the remaining $k - 2$ can land on the grid by Galois conjugacy, and do. The period-two construction we built to get past 6 was unnecessary and its values are superseded.
- **Refuted (ours).** That $k = 3$ and $k = 7$ admit no rational period-one symbol attaining $2k + 2$. That enumeration imposed $c^2 > 0$, the condition for a *symmetric* certificate, where $c^2 \neq 0$ suffices: c^2 is the product of the two spoke weights and only its vanishing is fatal. Six symbols at $k = 3$ and one at $k = 7$ survive, giving Theorems 24 and 27.
- **Refuted (ours).** The unconditional bound $Z(B^n) \geq D$ for cyclic covers, by $Z(P(8, 2)) = 5 < 6$. Only the conditional form $Z(B^n) \geq \nu$ for an attained ν survives.
- **Refuted (ours).** The single-threshold form $n_0(k) = 5k - 2$, by $Z(P(29, 6)) = 12$; and any bound of the form $Z \leq 2(\gcd(n, j) + \gcd(n, k))$, by $Z(P(11, 3)) = 7 > 4$.
- **Refuted (ours).** That the threshold for reaching the ceiling with a general matrix is $n \geq (2k + 2)^2/5$. Spending the gauge gives (3), $n \geq (k + 1)(2k + 3)/3$, which is larger; the earlier figure led us to test $k = 6$ at $n = 27, 29, 31$, all below the true threshold of 35.
- **Refuted (ours).** That $N(k)$ in Conjecture 51 is $O(k)$. The saturation data alone suggests $4k + 3$, but (3) forces $N(k) \geq \lceil (k + 1)(2k + 3)/3 \rceil$, which is quadratic.
- **Refined (ours).** We earlier recorded as refuted the claim that the monodromy is symplectic, on the grounds that the *standard* form is not preserved. That is true and was the wrong conclusion to draw. With the docking pattern fixed there is a unique invariant skew form B , and it is nondegenerate, so tile products are symplectic with respect to B (Observation 60). The earlier statement — that the space of invariant skew forms for a single product has dimension $k + 1$, so eigenvalues come in reciprocal pairs — remains correct; it is the family, not the individual, that pins B down.
- **Observation, not proved.** Observations 60 and 61: the invariant form and the full-dimensionality of the reachable set, verified for $2 \leq k \leq 6$ but not proved. These are what reduce the existence of identity tiles for all k to a controllability question in $Sp(2k + 2, \mathbb{R})$, stated in Section 11. We do not claim that reduction settles the question: subsemigroups with nonempty interior in $Sp(2n, \mathbb{R})$ need not be the whole group.
- **Inconclusive, not negative (ours).** The per- n search of Section 11 at $k = 6$ returned nothing at $n = 41, 43, 45, 47, 49, 51$, and the tile search at $k = 5$ nothing at $41 \leq \ell \leq 45$ — the latter now explained: the first $k = 5$ tile is at $\ell = 54$, so that search was below the range where tiles exist. Both ran at a restart budget that we have since seen condemn a length that does solve: $\ell = 21$ at $k = 3$ stalled at 5.6×10^{-2} on four restarts and produced a tile at 1.4×10^{-14} on twenty-five. We therefore record these as searches we did not push far enough, not as evidence about $k = 5$ or $k = 6$. The one regularity we have found is that every length eventually solved first appeared at a residual of 10^{-2} or below, and the $k = 5$ and $k = 6$ attempts sit at 10^{-1} .
- **Withdrawn as unreliable (ours).** The eigenvalue-minimising search whose plateaus we previously reported as evidence that the ceiling is unreachable at various (n, k) . It gave maximum nullity 7 at $P(20, 3)$, where 8 is attained — by Theorem 24 and, in the symmetric class, by the method of Section 11. It collapses onto the degenerate strata (all spokes zero,

or all edges zero). Its negative results should not be read as evidence, and the conclusion we drew from them, that exactness for $k \geq 6$ was unlikely, was not supported.

- **Open.** Conjecture 51 for $k \geq 6$, where we have no tiles at all and the count gives only $\ell \geq 54$. At $k = 5$ the conjecture now holds for $n \geq 162$ (Theorem 64) but not below it: we have 21 of the 54 lengths $[54, 108)$ would require, and $54 \leq n < 162$ is settled only at the 64 values that are sums of them. Also whether $M = Z$ in general, and the rank law. We expect $M(P(10, 2)) < Z(P(10, 2)) = 6$, a strict gap, since the search fails there even though the reachable dimension is saturated, but we do not prove it and no (n, k) is proved to have $M(P(n, k)) < 2k + 2$.
- **Superseded (ours).** The claim that the period- d method “cannot settle prime n at all” and that a different construction would be needed there. That is correct for the period- d method and is now a theorem (Theorem 38), but it is no longer a limitation of the paper: Section 11 settles prime n for $k = 3$.
- **Not determined.** Whether Theorem 7 is attained for $k \geq 3$; whether the degeneracy of Section 10 at $n = 48, 54$ is essential.

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